



AI Coding Best Practices: The Lovable Developer Guidebook

AI Coding Best Practices: The Lovable Developer Guidebook aims to help full-stack developers and AI prompt engineers harness Lovable – an AI-powered app builder – effectively. We'll cover eight chapters with practical guidance, prompt examples, do's and don'ts, and advanced tips. Whether you're building a simple landing page or a complex SaaS, this guide will help you collaborate with AI as a "lovable" co-developer.

1. Introduction to AI-Aided Development

Lovable is an AI-driven development platform that lets you build full-stack web applications by **describing what you want in natural language** ¹ ² . Instead of manually coding every component or setting up infrastructure, you enter prompts (instructions) and Lovable's AI (powered by large language models) writes the code, configures databases, and even deploys the app for you ² ³ . This approach drastically accelerates development – non-programmers can create apps without coding, and experienced developers can delegate boilerplate tasks to the AI.

Key capabilities of Lovable's AI-aided platform include: front-end UI generation, database integration (via Supabase), user authentication, backend functions, real-time collaboration, and one-click cloud deployment ⁴ ⁵ . For example, you could prompt: *"Build a task tracker with a React frontend, Supabase backend, and user login."* Lovable will scaffold the project with those features. The AI acts like a smart pair-programmer who understands high-level requests and writes the underlying code.

How AI-aided development differs from traditional coding is in the workflow and skills required. Instead of writing low-level code line by line, you focus on **communicating requirements clearly to the AI**. This is often called *prompt engineering* – crafting precise instructions so the AI produces the desired outcome. It's a new skill for developers: think of instructing a very literal junior developer. The AI has no intuition beyond your prompt, so success depends on how well you specify the behavior and constraints ⁶ ⁷ . (We'll delve into prompt strategies in Chapter 2.)

That said, Lovable doesn't eliminate coding completely. It's a **flexible environment**: you can inspect and edit the generated code at any time (especially with *Dev Mode*, which exposes the full codebase in a built-in editor ⁸). This means experienced developers can fine-tune complex logic or optimize performance manually when needed. You're not locked in – every Lovable project is backed by a Git repository, so you have version control and can even export or self-host the code ⁹ ¹⁰ .

Benefits of AI-aided development with Lovable include:

- **Speed from Idea to App:** Go from a vague concept to a working prototype in minutes. For example, simply prompting "Create a landing page for a space tourism agency" can generate a multi-section, responsive page with placeholder content, ready for customization ¹¹ ¹² . This lets you validate ideas extremely fast.

- **Natural Language Interface:** Describe features or fixes in plain English. You might say: “Add a signup form with name, email, password fields and validate the email format,” and the AI will implement it. This lowers the barrier to development for those who aren’t fluent in programming syntax.
- **Full-Stack Coverage:** Lovable integrates front-end components, backend services, and deployment. For instance, if you prompt the AI to “Save submitted contact form data to a database,” it will create a Supabase table and the necessary API/edge function to insert the data securely ¹³ ¹⁴ . Deployment is just a click – publish to a shareable URL or connect a custom domain when ready ¹⁵ ¹⁶ .
- **Collaboration and Iteration:** AI-aided development is highly iterative. You can continuously refine your app via chat-based interactions. In Lovable’s **Chat Mode**, the AI is aware of your project’s context (files, schema, recent changes) and can discuss solutions or brainstorm with you ¹⁷ ¹⁸ . This is like having a project-aware assistant: you describe an issue or new feature, and the AI proposes and implements changes. Real-time collaboration is supported too – teammates can join your Lovable workspace and see edits live, with multi-cursor editing and shared history ¹⁹ ²⁰ .

Despite these advantages, **AI-aided development introduces new best practices**. You must guide the AI deliberately. A common pitfall is to jump in without a plan: “Users start building before they’ve clarified what they’re building,” leading to meandering prompts and confused AI output ²¹ . It’s crucial to outline your idea first – we recommend writing a short project brief or list of user stories *before* prompting (even if it’s just in a notebook or a separate document) ²² ²³ . This clarity will save you from “errors on top of errors” caused by an evolving vision ²⁴ . Many successful Lovable builders actually **start outside Lovable**: they record themselves describing the idea or draft a Product Requirements Document, use ChatGPT to refine it, and enter Lovable with a solid plan ²⁵ ²⁶ .

In summary, Lovable enables a *high-level, design-first approach* to development: you articulate what to build and the AI figures out how. Embrace this shift by focusing on **what** the app should do (functionality, UX, constraints) and letting the AI handle **how** to do it in code. The following chapters will teach you how to communicate with the AI effectively, troubleshoot when things go wrong, integrate external services, and follow best practices so your AI-assisted project remains *lovable* in quality and maintainability.

2. Prompt Engineering Strategies & Frameworks

Prompt engineering is the art of communicating your intent to the AI in a way that yields correct and helpful code. In Lovable, **prompts are your primary tool** – they’re like instructions you give to a very literal-minded assistant ⁶ . Mastering prompt strategy can dramatically improve the AI’s output ²⁷ . This chapter introduces frameworks and tips for crafting effective prompts.

2.1 The C.L.E.A.R. Framework: A handy guideline for prompts is to keep them CLEAR – **Concise, Logical, Explicit, Adaptive, and Reflective** ²⁸ ²⁹ :

- **Concise:** Get to the point. Provide enough detail, but avoid unnecessary fluff or ambiguity. For example, instead of “Could you maybe write something about a science topic?”, say “Write a 200-word summary of the effects of climate change on coastal cities.” (The latter is precise about length, topic, and scope) ³⁰ ³¹ . Unfocused prompts waste tokens and can confuse the model. Aim for brevity *with clarity* – every sentence should inform the task.
- **Logical:** Structure your instructions in a step-by-step, coherent manner. Break complex requests into ordered steps or bullet points ³² . For instance, “Build a user sign-up feature and also show some stats”

is vague and conflates tasks. A logical prompt would be: “1. Implement email/password sign-up using Supabase Auth. 2. After successful sign-up, display a dashboard with a user count statistic.” This way, the AI knows to handle sign-up first, then dashboard, in sequence ³³.

- **Explicit:** State exactly what you want (and what you **don’t** want). Never assume the AI will fill in the blanks. If you need a specific library or format, mention it. For example, “Tell me about dogs” is too open-ended, whereas “List 5 unique facts about Golden Retrievers in bullet points.” gives clear directions on content and format ³⁴. Likewise, if certain files shouldn’t be touched or certain styles must be used, spell that out (e.g. “Do not modify `Header.jsx`” or “Use Tailwind CSS for all styling”) ³⁵. Treat the AI as if it has no common sense – because in a way, it doesn’t.
- **Adaptive:** Be prepared to iterate and refine. After the AI responds, review the output critically. If it missed something, adapt your next prompt to clarify or correct. You can even ask the AI how to improve your prompt (this is called *meta-prompting*, see section 2.3). For example, “The login function you wrote is missing validation for email format. Please update it to include proper validation.” By giving feedback or follow-up instructions, you guide the AI closer to your goal ²⁹ ³⁶. Lovable’s chat interface supports this iterative dialogue – use it to tweak the behavior.
- **Reflective:** Learn from each AI interaction to improve future prompts. After a successful change, note what prompt phrasing worked well. After a poor result, analyze why. You can even have the AI summarize what it did (“Summarize the changes you just made and why”) to ensure understanding. Being reflective might involve asking the AI to explain its code so you can spot misunderstandings ³⁷. Over time, this builds your intuition for which prompting strategies yield the best outcomes.

2.2 Structured vs. Conversational Prompting: Depending on the scenario, you may use a more structured “form” for prompts or a conversational style:

- **Structured “Training Wheels” Prompts** – For complex tasks or when you’re new, it helps to organize the prompt into labeled sections like **Context, Task, Guidelines, Constraints** ³⁸ ³⁹. This format forces you to provide all relevant info. For example:

Context: “You are an expert full-stack developer using Lovable.”

Task: “Build a secure login page with email/password using Supabase for auth.”

Guidelines: “Use a minimalist UI with Tailwind CSS. Provide clear error messages for failed login attempts.”

Constraints: “Only modify the `LoginPage` component file; do not affect other pages. Do not use any paid libraries.” ⁴⁰ ⁴¹

This explicitly tells the AI its role, what to do, stylistic preferences, and limits. Such a prompt might yield a well-structured `LoginPage` component with comments and error handling. Structured prompts are excellent for major features or when you want to be absolutely sure the AI has all necessary context.

- **Conversational Prompts** – Once you’re comfortable, you can write prompts more naturally, like talking to a colleague ⁴². You might say: “Let’s add a profile picture upload feature. We need a form with an image file input and a submit button. On submit, store the image in Supabase Storage and update the user’s profile. Handle errors (like file too large) gracefully.” ⁴³ This is less formally segmented, but still clear and ordered. In conversational style, you rely on context from previous messages or the project state. It’s important to **still be precise and complete** – even without section headers, you

can break the prompt into paragraphs or bullets for clarity ⁴⁴. Conversational prompting is great for quick iterative changes or in the middle of a chat-driven development session.

Tip: In either style, consider formatting your prompt content to guide the AI's output. For example, if you want an **ordered list** of steps in the response, enumerate them in your prompt (1., 2., 3.) ⁴⁵ ⁴⁶. If you want output code separated from explanation, explicitly say so (e.g. "First explain the approach, then provide the code.").

2.3 Levels of Prompting – Meta and Reverse Meta: Beyond basic instructions, Lovable's AI can help improve itself:

- **Meta-Prompting (AI helping with prompts):** You can ask the AI to critique or refine your prompt. For instance: "Review my last prompt and suggest how to make it more specific and clear." or "Rewrite this prompt to follow best practices: 'Create a secure login page...'" ⁴⁷ ⁴⁸. The AI might respond with a cleaned-up prompt that includes details you omitted. This is a powerful way to leverage the AI's perspective to improve your communication. In Chat Mode, meta-prompts are safe to use because they won't directly change your code – you're just asking for advice ⁴⁹.
- **Reverse Meta-Prompting (AI as a documenter):** This means using AI to document or verify code behavior, essentially turning the AI's lens back on the project. For example, "Explain in simple terms what the `calculateStats()` function does and list any edge cases it doesn't cover." The AI will analyze your code and produce a description ⁵⁰ ⁵¹. This helps ensure the AI (and you) fully understand the current code, and it can reveal misunderstandings. Another reverse-meta trick: after a series of changes, ask "Summarize the current project structure and any remaining TODOs." This can surface hidden issues or remind the AI of context if the conversation is long. Essentially, you use the AI as a second pair of eyes on the code.

2.4 Context is King: Large Language Models (LLMs) have no knowledge of your project beyond what you provide. They don't "see" the UI or "know" your app's purpose unless you tell them ⁵². Always **provide context** in prompts: mention the page or component name, the user story, or relevant schema. For example, instead of "Add a comments section," say "In the `PostDetail` page, add a comments section where logged-in users can add a comment to the current post (identified by post ID)." This ensures the AI knows which file and data model to work with. The more project-specific context you include, the less the AI has to assume (which reduces mistakes).

Lovable helps with context through features like **Knowledge Files** – a special markdown file where you can write project background, requirements, data models, etc. The content of the Knowledge file is automatically included with every prompt as supplemental context ⁵³ ⁵⁴. For example, if your knowledge file describes user roles and permissions, the AI will consider that when you prompt about features ("Investors can see dashboard stats, Admins cannot" etc.). It's a best practice to maintain the knowledge file as your project's single source of truth (more on this in Chapter 6). Using knowledge files means you don't have to repeat basic info in every prompt – it **saves token budget and keeps the AI informed persistently** ⁵⁵.

2.5 Using Lovable's Modes Intentionally: Lovable has different modes (Design/Default Mode vs. Chat Mode vs. Visual Edit) – use them for what they're best at ⁵⁶ ⁵⁷ :

- **Chat Mode** is ideal for brainstorming, debugging, and high-level planning. Example: *"Why might the login function be returning undefined? What could be the cause?"* – In Chat mode, the AI will analyze the code without immediately editing it, acting like a tutor or debugger ⁵⁸ ⁵⁹ . Use Chat mode to discuss solutions before committing them. It's also great for non-destructive tasks like code reviews (e.g., *"Audit the codebase for any anti-patterns or overly complex areas."*) or generating unit test ideas.
- **Default Edit Mode** is when the AI directly writes or edits code. This is triggered by prompts in the main editor (outside Chat mode). Use this when you are confident in the change you want. Often, you might workshop an approach in Chat, then switch to Default mode to implement it: *"Apply the fix by only changing the login() function to include the new validation."* ⁶⁰ . Because Default mode edits can consume your message credits (especially if an edit fails and you retry), it's wise to do complex problem-solving in Chat mode first ⁶¹ . Think of Chat mode as planning/drafting, and Default mode as execution.
- **Visual Edit Mode** allows you to select UI elements on a live preview and modify styles or content through natural language. This is perfect for quick UI tweaks: you click an element (say a button) and type *"Make this button blue with 16px padding"*. The change is applied instantly without writing a manual prompt in the chat ⁶² . Visual edits don't cost AI credits and can be undone, so they're a fast, free way to fine-tune the interface. Use visual editing for layout adjustments, CSS tuning, or text changes – it's often easier than describing the coordinates or class names via text prompt. (Under the hood, Lovable generates the appropriate Tailwind classes or style objects for you.)

By combining modes, you can adopt a **efficient workflow**: Plan in Chat -> Implement in Code -> Touch-up in Visual editor. For example, if starting a new feature, you might chat: *"I want to add a blog section. What components and data model will I need?"* The AI gives an outline. Then you exit chat and prompt in Default mode: *"Create a BlogPost page and Supabase table for blog posts as discussed."* After the code is generated, you switch to visual mode to tweak spacing or colors. Each mode has its strengths – **use the right tool for each part of the job** ⁶³ ⁵⁷ .

2.6 Additional Prompting Tips and Tricks:

- **Examples and References:** If you have a target style or reference implementation, provide it. This can be a link to documentation or even an image. For instance: *"Make the login page UI similar to the design in the attached screenshot."* Lovable allows image uploads with prompts, so you can literally show the AI what you want ⁶⁴ ⁶⁵ . Another way is referencing known patterns: *"Use a Chakra UI <Modal> for the settings dialog."* The AI has knowledge of many libraries and will emulate their usage. Including code snippets as examples can also help (e.g. *"Here is how I fetch data in another page: <code>... Use a similar approach in the new page."*). Concrete examples set a clear expectation ⁶⁶ ⁶⁷ .
- **Constraints and Guardrails:** We touched on explicitly telling the AI what not to do. One pattern is a **"do not modify X"** instruction if you fear collateral changes. For example: *"Add the search feature to the ProductsPage without altering the Header or global layout."* This is essentially "locking" those files via prompt ³⁵ . The AI will then confine its edits to the intended scope. Another guardrail is

cautioning the AI about fragile areas: *"The payment processing code is critical, so be careful not to break it while adding this new feature."* This primes the AI to be more conservative around that section ⁶⁸ ⁶⁹. It's not foolproof, but it reduces reckless changes.

- **Tone and Politeness:** While the AI doesn't have feelings, writing prompts in a polite, calm tone can indirectly improve outcomes – it encourages you to be more descriptive. Saying "please" and explaining the request clearly (as you would to a human) often results in more coherent responses ⁷⁰. For example, *"Please refactor this function for clarity, without changing its functionality."* is both courteous and precise. The AI will focus on clarity and keep behavior unchanged, as asked.
- **Avoiding Ambiguity:** Watch out for words like "it" or "they" with unclear antecedents, or requests that could be interpreted in multiple ways. Instead of *"Fix it so it doesn't break if no data,"* say *"Handle the case where the users array is empty – e.g., show a message 'No users found' instead of throwing an error."* Ambiguity is a common cause of "AI did something weird." Always specify the scenario and desired behavior exactly to avoid misunderstandings ⁷¹ ⁷².
- **Incremental Changes:** It's usually best to ask for one feature or fix at a time (or a logical group of small changes), rather than a huge list of modifications in one prompt. For instance, break a large task into a series of prompts: *"First, create the UI for the profile page."* After that's done, *"Now add the logic to save profile changes to the database."* This stepwise approach helps isolate issues and makes it easier to pinpoint which prompt introduced a bug ⁷³. As a rule of thumb, *avoid trying to implement 5 things at once* in one prompt ⁷³. Smaller prompts keep the AI (and you) focused.
- **Model Limitations:** Remember that Lovable's AI has knowledge cut off at a certain date and may not know about the latest libraries or events after that. If you need something current (say a brand-new API), you might have to provide documentation or code examples for it. Also, if the AI starts hallucinating (making up functions or APIs that don't exist), it's a sign to either provide factual references or simplify the request ⁷⁴ ⁷⁵. Use higher-level models (like GPT-4) for complex reasoning and ensure your prompt gives any non-public info (like special formulas or business rules) because the AI won't know those unless told.

In summary, **good prompting is about clarity and control**. Start with a clear goal, give necessary context, break down the tasks, and specify the details of output you expect. Then iterate – evaluate the AI's output and refine your next prompt accordingly. With practice, you'll develop a feel for instructing Lovable's AI effectively. Think of it as increasingly seamless pair-programming: you focus on *what* needs to happen, and the AI figures out *how*, guided by your well-crafted prompts.

3. Debugging Workflows & Troubleshooting Prompts

Building with AI can feel magical when everything works – but what about when something goes wrong? Perhaps the app crashes with an error, or the AI isn't following your instructions, or a feature just isn't behaving as expected. **Debugging in an AI-driven workflow** involves a mix of conventional debugging (checking error logs, code, etc.) and conversing with the AI to diagnose and fix issues. This chapter outlines a structured approach to troubleshoot problems and refine prompts when Lovable doesn't get it right.

3.1 Quick Triage: Use Automated Fixes First. Lovable provides a **“Try to Fix” button** whenever your project encounters a runtime error. This is your first line of defense ⁷⁶. Clicking it prompts the AI to scan the error log and attempt a direct fix. For simple issues (a typo, missing import, etc.), this often resolves it immediately ⁷⁷. Always try this one-click fix – it’s fast and costs nothing. However, if the AI repeatedly enters a **“Try to Fix” error loop** (fixing one thing only to break another, over and over), it’s time for deeper debugging ⁷⁸. Don’t keep hitting fix blindly; move on to investigation.

3.2 Diagnose Unexpected Behavior. Not all bugs throw errors – sometimes the app just doesn’t do what you intended (e.g., a button click does nothing). When you notice **unexpected behavior**, take a step-by-step approach ⁷⁹:

- **Re-read your original prompt.** Ensure that what you asked aligns with what you wanted. Often, clarifying your prompt can solve misbehavior. For example, if *“logout on idle”* ended up logging out users immediately, you may realize you didn’t specify an idle duration – a prompt refinement can fix that.
- **Check each component or function in isolation.** Break down the feature to find where it deviates from expectations ⁸⁰. For instance, if a form isn’t saving data, verify the form submission event triggers, then whether the API call is made, then whether the database received data. This may involve using browser dev tools or console logs (which you can also ask the AI to insert for you).
- **Add visuals or logs for clues.** If it’s a UI issue, take a screenshot of the glitch and even **upload it in Chat Mode**. You could ask: *“The attached screenshot shows the layout issue – the sidebar is missing. What might be wrong?”* Visual context can help the AI pinpoint problems (like a missing CSS class) ⁸¹. For logic issues, copy any error messages or relevant logs and feed them to the AI in chat, e.g., *“Here’s the error from the console: [paste]. What does this suggest?”*

Lovable’s Chat Mode is **project-aware**, meaning the AI can often draw on the context of your code to help debug. You might simply prompt: *“Something’s off – the page goes blank after I submit the form. Can you walk me through what might be happening and what was attempted so far?”* This invites the AI to analyze the current project state and recent changes ⁸². It often provides a narrative of what likely went wrong.

3.3 Write More Effective Follow-up Prompts. If a bug arises from a prompt misunderstanding, your next step is to write a *better prompt*. A good strategy is to **restate the goal, step by step**, so the AI can’t misinterpret. For example, suppose you initially prompted *“Add search to the user list”* and it isn’t working. A more effective prompt might be structured like ⁸³:

1. **Project Overview:** We have a page listing users. I want to add a search bar to filter this list.
2. **Page Structure:** The page is `UserList.jsx` with a list of users rendered. It should have an input field at the top for search.
3. **Search Logic:** Typing in the search box should instantly filter the shown users by name or email (client-side filter).
4. **No External API:** Do not call any external API; filter the already loaded users array in state.
5. **Keep Everything Else Intact:** No changes to how users are fetched or other page parts.

This kind of prompt gives a clear blueprint (it's essentially a mini-PRD for the feature) ⁸⁴ . By explicitly listing requirements and constraints, you guide the AI to a correct solution. **If a bug was caused by a vague prompt, this approach will prevent recurrence.** Remember: clear and structured prompts are as useful for debugging as they are for initial development.

3.4 Build (or Re-build) in Logical Order. When a project grows complex, errors often creep in if things were built in a haphazard order. A recommended **order of implementation** in Lovable is ⁸⁵ :

1. **Layout & Pages:** Ensure the navigation and page components render properly.
2. **Connect the Database:** Integrate Supabase (or backend) once the frontend skeleton is ready.
3. **Authentication:** Set up user login/roles early, as it can affect subsequent logic (e.g., who can see what).
4. **Feature Logic:** Add core features one by one, testing each in isolation.
5. **Polish & Edge Cases:** Only after the happy paths work, add validations, loading states, etc.

Following this sequence prevents many issues. For example, if you connect the database *after* building UI, Lovable can auto-generate the needed schema without conflicts ⁸⁶ ⁸⁷ . Conversely, if you connected Supabase first and then heavily changed the data model via prompts, your Supabase schema might become inconsistent. (In fact, *reverting* to a previous version of code won't revert the database – a common source of bugs if not careful ⁸⁸ ⁸⁹ .) So, to troubleshoot structural issues, consider if something was built out of order. You might need to “reset” or refactor parts to follow a logical flow. Lovable's **Remix** feature can help here (it lets you fork your project state and rebuild parts cleanly – we'll cover this soon) ⁹⁰ ⁹¹ .

3.5 Leverage Chat Mode for Debugging Dialogues. Chat Mode truly shines during debugging. Use it to have a conversation with the AI about the problem. Here's a reliable workflow when you're stuck in an error loop or mysterious bug:

1. **Switch to Chat Mode** (to avoid making unintended code changes during analysis) ⁹² .
2. **Present the issue clearly:** e.g., *“After adding the notification feature, emails aren't sending. I expected an email when a task is overdue, but nothing happens.”* ⁹³ Include relevant logs or error messages: *“The server log shows: ‘permission denied when trying to send email’.”* ⁹⁴ .
3. **Ask targeted questions:** *“What could cause this permission error when sending email?”* or *“The AI tried solution X and Y already (list them). What else can we try?”* ⁹⁵ ⁹⁶ .
4. **Let the AI analyze and suggest fixes.** It might deduce, for example, that the email API key was not configured, or a database rule is blocking the operation ⁹⁷ .
5. **Iterate interactively:** Based on the AI's suggestion, you could say, *“Okay, if the API key is missing, guide me to securely add it.”* This might involve adding a Supabase secret or using Lovable's **Add API Key** form (so the key isn't hard-coded – recall the security best practice ⁹⁸ ¹³).

Throughout, treat the AI like a colleague. Explain what you think the issue is, or ask it to explain its reasoning. A great tactic is to request a **step-by-step analysis**: *“Let's debug this together. First, analyze the error. Next, check what was tried. Then suggest a fix plan.”* The AI will follow that structure, which can reveal the root cause systematically ⁹⁹ ¹⁰⁰ .

3.6 Handling Persistent Errors: If an error keeps coming back in different guises, the root cause might not have been fully addressed. In such cases:

- **Ask the AI what it has already tried.** Literally: *“What solutions have we attempted so far for this error?”* The AI can list the fixes it applied (or you applied) in previous steps ⁹⁵. This helps avoid repeating the same approach and expecting different results.
- **Have the AI explain the error in plain language.** *“Explain why this error occurs, in simple terms.”* Sometimes the act of explanation uncovers a hidden detail – maybe the AI misunderstood a requirement, or you did. If the explanation doesn’t match your expectations, that’s a clue.
- **Consider an alternate implementation.** Ask: *“Given that this approach keeps failing, can we try a different strategy to achieve the goal?”* ⁹⁶. The AI might propose a workaround or a entirely new method that bypasses the problematic part. For example, if writing to the database directly is error-prone, maybe use an intermediate API or adjust the schema.
- **Rollback and redo in smaller steps.** In Lovable, you can restore a prior version of your project (via version history) to undo a tangled mess ¹⁰¹. If you do this, tell the AI: *“I’m reverting to before we added feature X. Let’s reintroduce it step-by-step.”* This “fresh slate” approach often succeeds because you can watch for the exact step where things go awry.

Lovable’s **Remix** feature (mentioned above) is helpful when a project feels “too far gone.” Remix creates a new project copy at the current state but without the connected database, allowing you to re-bind or reconstruct that part cleanly ¹⁰² ¹⁰³. Many users find that *rebuilding a feature the second time is much faster and cleaner*, since you know the pitfalls. Don’t hesitate to leverage remixing rather than spending hours in a dead end – it’s sometimes the wisest debugging move ⁹⁰ ⁹¹.

3.7 Common Problem Scenarios & Solutions:

- **“Nothing happens when I click the button” (UI action issue):** Ensure the onClick handler is correctly wired and not accidentally removed. You can prompt, *“Why does the Save button on ProfilePage do nothing on click?”* The AI might realize the function name changed or that an onSubmit wasn’t bound ⁷⁹. Solution: explicitly reconnect the handler (or re-add it if missing). In Visual Edit mode, you might quickly verify if the button has the right binding.
- **“The UI layout is broken or element disappeared”:** Possibly a CSS issue or a React key error. Use screenshots and ask the AI. If, say, a side panel vanished, the AI might recall that a condition hides it (e.g., if no data). It might tell you to check a conditional render. You can also use Chat Mode: *“The sidebar component isn’t showing up on mobile – here’s how it looks (screenshot). What could be wrong?”* It could suggest adding responsive classes or ensuring the component is imported properly ¹⁰⁴. Lovable’s AI might even suggest a fix like: *“Perhaps the sidebar is not included in the mobile layout component. Ensure it’s rendered or use responsive design classes.”*
- **“Stuck in an error loop”:** This is where every fix leads to a new error (often downstream of the initial issue). The best practice here is to break the cycle by switching context: go into Chat Mode and ask *“What is the root cause of these errors?”* rather than *“Fix it.”* ¹⁰⁵ ¹⁰⁶. For example, if you kept fixing

TypeScript types only to hit another type error, maybe the root cause was a wrong data model assumption. Asking for root cause analysis can surface that big-picture issue. Once identified, you can address it with a targeted prompt. The guide from the AI might be: *“It appears all these type errors stem from the `User` interface being mis-defined. Let’s correct that interface.”* Then you implement that correction.

- **“Feature not working right (no errors)”**: Often a logic or integration issue. For instance, notifications not sending, as earlier. Here, engage the AI in a hypothetical test: *“Walk me through what should happen when a task is overdue. Where could it be failing?”* The AI might prompt you to check if the serverless function was deployed, or if the cron job (if any) is running. This scenario is where logs are crucial – share any relevant log excerpt with the AI ⁹⁴. The AI deduced in our earlier example that the API key wasn’t set because of a “permission denied” log ⁹⁷. In another case, a silent failure might be due to an unhandled promise. The AI could suggest adding a `.catch` to see the error.
- **“Hallucinated or incorrect code”**: Sometimes the AI writes code that looks plausible but is wrong (e.g., using a wrong function name, or a library method that doesn’t exist). If you spot this, correct it explicitly: *“The AI used a function `getUserData()` that doesn’t exist. Use the existing `fetchUserData()` function instead.”* Also, this is a learning moment – your next prompt should remind the AI of actual project entities. This is where having your **Lovable Knowledge** file updated (with actual function names, data schemas) helps prevent hallucinations ¹⁰⁷ ⁷⁵. In Chat Mode, you can also just ask: *“Is this code valid? It uses X – is that defined?”* The AI will often double-check and admit mistakes.

3.8 Involve Humans When Needed: Despite the AI’s help, sometimes you’ll need a human perspective. Lovable has a community Discord and support channels ¹⁰⁸. Knowing when to ask for help is a skill too. If you’ve spent a long time and the AI’s suggestions are looping, consider reaching out with a clear summary of the issue (and what you and the AI have tried). Often a fresh set of eyes (or someone who built a similar feature) will spot the issue quickly ¹⁰⁹. The AI itself might recommend seeking human help if the problem is out of scope or if it’s hitting its own limits ¹¹⁰.

3.9 Preventative Debugging – Prompting for Quality: It’s better to catch issues early than fix them late. You can use prompts proactively to **audit and test your project**. For example, use the **“Full Codebase Audit” prompt** (from Chapter 2’s examples) to have the AI review the entire project structure for any misplaced logic or bad practices ¹¹¹ ¹¹². The AI might output a checklist of improvements, effectively acting like a code reviewer ¹¹³ ¹¹⁴. You can then address those before they become user-facing bugs.

Similarly, ask the AI to **generate test scenarios**: *“List some edge cases for the payment feature that we should test.”* It could respond with cases like “network failure during payment, invalid card details, user not logged in while purchasing” etc. You can then manually or with code test those scenarios. While Lovable doesn’t automatically generate full test suites yet, using the AI to think through edge cases is extremely helpful in hardening your app ¹⁰⁰.

3.10 Example – Debugging a Recurring Error: Let's tie it together with a realistic scenario. Say you have an app where users can upload images, but sometimes the upload fails with a generic error. You've tried "Try to Fix" and the AI changed some code, but it still fails. Here's how you might proceed:

- In Chat Mode: "The image upload occasionally fails with error code 413. What could be the cause?" The AI explains 413 means payload too large. It suggests perhaps the image is too big or not being chunked ⁵⁹.
- You realize you didn't set any limit. So you prompt: "Add client-side validation to prevent uploading images larger than 5MB. Also, adjust the server function to handle large files (maybe compress or reject with a clear message)." This prompt is clear about what to do and the threshold. The AI implements it.
- You then say: "If the error happens again, log the image size and maybe thumbnail it." The AI updates the code to log and attempt thumbnail generation.
- After deployment, things are smooth. **By iteratively conversing**, you fixed the issue. Additionally, you learned to specify file size limits proactively next time.

Throughout the debugging journey, remember you are **collaborating with the AI**. It has read tons of code and can offer valuable insights, but *you* steer the investigation. Verify each AI-proposed change (read the diff! ask yourself if it makes sense). Use version control to your advantage – if a fix makes things worse, you can revert to a previous commit (the Lovable interface shows version history; pin good versions so you can always fall back ¹¹⁵ ¹¹⁶). Debugging with AI is a skill that improves with practice. Over time, you'll develop an intuition for diagnosing issues and you'll train the AI (through better prompts) to avoid them in the first place.

In the next chapter, we'll discuss system design and optimization – which often pre-empts many bugs by structuring the project well. But when bugs do happen, use the structured workflows discussed here. With a clear head and a clear prompt, no bug will stay unsolved for long!

4. System Design, Optimization, and Deployment

Building a robust application is not just about writing code that works – it's about designing the system architecture thoughtfully, optimizing for performance and security, and deploying it reliably. In the context of AI-assisted development, **system design** involves guiding the AI to implement a clean architecture, while **optimization** means fine-tuning the app's performance and resource usage (often with the AI's help), and **deployment** covers publishing your Lovable app to the world. This chapter provides best practices in these areas.

4.1 Planning the System Design (Before You Prompt): A bit of upfront planning goes a long way. Before diving into prompts, outline your app's key entities and how they interact. Many seasoned Lovable users **start by writing a short spec or list of features**, even feeding that to the AI as context. For example, you might maintain a bullet list like:

- *Users* can sign up, create projects, and add tasks.
- Each *Project* has a name, description, and list of tasks.
- *Tasks* have title, due date, status, and belong to a project. Only project owners can mark tasks complete.

You can include such details in your **Knowledge File** – Lovable’s living project documentation ¹¹⁷ ¹¹⁸ . The knowledge file can also capture non-functional requirements (e.g., “*should support 10k users*,” “*must comply with GDPR for data*”), which influence design decisions. By populating this file early (either manually or by asking the AI to generate a draft based on your idea) ¹¹⁹ , you ensure the AI has a strong context about your system’s intended design.

Also, consider high-level *approach* before building. In Chapter 1, we mentioned two styles: **Front-end first vs. Back-end first**. For many projects, starting with the front-end (UI and mock data) is wise ¹²⁰ ¹²¹ . It lets you iterate on user experience and catch design issues before entangling real databases. For example, you can prompt: “*Create a dashboard page with sample data (no real database yet) to show the layout.*” Then once it looks right, connect Supabase: “*Now connect this page to Supabase – the statistics should come from a stats table.*” This avoids dealing with SQL errors or auth issues while you’re still figuring out layout ¹²² . Lovable makes it easy to plug in Supabase when ready (via the Integrations menu, then generating schema as needed) ¹²³ ¹²⁴ . The alternative **Back-to-Front** approach (database first) can be efficient for experienced devs, but it demands careful planning of the schema up front ¹²⁵ . If you go this route, thoroughly define your DB schema in the knowledge file and perhaps have the AI generate migration scripts for review.

Avoid the “building before planning” trap by doing the above. A common result of no planning is projects that veer off track or become “too far gone to fix” without major refactoring ²⁴ . If you do end up in that situation, Lovable’s **Remix** (fork) feature is a lifeline to re-think system design: you can create a fresh project copy and re-align it with better architecture, without losing the old version for reference ⁹⁰ ⁹¹ . Of course, prevention is better – plan first!

4.2 Enforcing a Clean Architecture via Prompts: When working with AI, you sometimes need to explicitly ask for good architecture. The AI doesn’t automatically know, for example, how you want to separate concerns. Here are some prompt tactics:

- **Ask for an architecture audit.** If your project has grown, you can use a prompt like the **Codebase Audit** example: “*Perform a comprehensive audit of the entire codebase to check if the architecture is clean, modular, and optimized. Identify any files or logic in the wrong place, any overly coupled code, and any violations of best practices. Provide recommendations without changing code.*” ¹²⁶ ¹¹² . The AI might come back with something like: “The API calls are made directly in UI components (e.g., in `ProjectList` component) – consider moving them to a separate module or hook to separate data fetching from presentation ¹²⁷ . Some utility functions in `App.jsx` could be moved to a `utils` file ¹²⁸ . The task toggling logic is duplicated – could be refactored into one place,” etc. This gives you a roadmap to refactor for better system design.
- **Encourage separation of concerns in prompts.** For instance, when prompting, specify layers: “*Generate a new React context `ProjectDataContext` to handle data fetching for projects, and use it in the `ProjectList` component instead of fetching inside the component.*” This tells the AI exactly how to organize responsibilities ¹²⁷ . Another example: “*Create an Edge Function for sending emails (server side) and have the client call that instead of calling the email API directly from the front-end.*” By doing so, you guide the AI to a **distributed architecture** where appropriate – leveraging Supabase Edge Functions for backend logic (so you don’t expose API keys on the front-end) ¹³ .
- **Use Lovable’s file locking trick.** If you want to ensure the AI doesn’t cram logic where it doesn’t belong, explicitly instruct it: “*Do not put any database code in the React components; all DB access*

should go through Supabase or serverless functions.” or “Keep authentication checks in the backend only.” This acts as a constraint so the AI doesn’t, say, stick a secret API call in a client-side file (which would be a security flaw) ¹²⁹ ¹³⁰ .

- **Design for roles and modularity.** If your app has multiple user roles (Admin, Investor, User, etc.), plan out what parts of the system each role touches. Then prompt accordingly: *“Create separate components or views for admin-specific functions, to avoid intermixing admin logic in user components.”* Also explicitly mention role contexts in prompts: *“As an Investor (role), I should be able to see X but not edit it.”* ¹³¹ ¹³² . This reduces the chance that the AI will inadvertently give access where it shouldn’t.

By guiding the AI with these architectural instructions, you enforce a structure that’s easier to maintain. Remember, the AI is excellent at following patterns – if you set a pattern early (e.g., “all API calls via `api.js` module”), it will generally stick to it throughout the project unless told otherwise.

4.3 Performance Optimization Checks: Performance bottlenecks can emerge as your app scales or adds more data. Lovable’s AI can assist in identifying and fixing them:

- **Run a Performance Audit via prompt.** Similar to the code audit, ask: *“The app feels sluggish with large data. Analyze the project for performance bottlenecks and suggest optimizations (caching, avoiding re-renders, etc.), without changing code yet.”* ¹³³ ¹³⁴ . The AI will examine things like data fetching patterns, heavy computations on the main thread, image sizes, and so on. For example, it might find: *“`ProjectList` fetches data on every render – consider caching results or using context to avoid duplicate fetches ¹³⁵ . `TaskItem` component isn’t memoized and re-renders too often – consider using `React.memo` to prevent unnecessary re-renders ¹³⁶ . There’s a 2MB image being loaded – that’s large; compress it for faster loading ¹³⁷ .”* Each point provides a clear action to take (and you can prompt the AI to implement each). By heeding these recommendations, you can significantly improve responsiveness ¹³⁸ .
- **Request specific optimizations:** If you already suspect an issue (say a slow page), you can directly prompt: *“Optimize the rendering of the dashboard – it currently lags. Perhaps use pagination or lazy loading for the list of 1000 items.”* The AI might implement an infinite scroll or pagination component as suggested. Or *“Implement caching for the project data using a React context or state, so we don’t re-fetch on every navigation.”* ¹³⁵ ¹³⁹ . This way you target known hotspots.
- **Monitor and measure:** While the AI can guess optimizations, *you* should measure performance (e.g., using Chrome DevTools timeline, or simple `console.time` logs around functions) and feed those findings back. For instance: *“I profiled and found the `generateReport()` function takes 3 seconds. Can we optimize it?”* The AI might suggest breaking it into web workers or optimizing the loop. Always verify after changes – did the load time drop? If not, iterate further.
- **Use better algorithms or data structures:** Sometimes the AI’s initial code isn’t the most efficient (LLMs might write $O(n^2)$ algorithms for simplicity). If you notice slow behavior on large inputs, prompt the AI to improve complexity: *“Refactor this function to handle up to 10,000 records efficiently (it’s currently slow). Maybe use a Map for lookups instead of nested loops.”* The AI should then apply a more optimal approach. It’s particularly useful to mention known methods (like “use binary search” or “apply debouncing to this search input”) – the AI will incorporate those patterns.

In one case, a user had an AI build an image processing tool that was slow; they asked the AI to analyze memory usage and CPU usage, and it realized images were processed on the main thread. The AI then suggested moving heavy computations to a Web Worker (which Lovable supports through its code editor). This kind of high-level insight can come directly from asking the AI *why* something is slow and *how* to fix it.

4.4 Security Considerations in System Design: Security should be woven into your design from the start. Lovable integrates with Supabase, which offers security features like Row-Level Security (RLS) policies for databases. Key practices include:

- **Never expose secrets in front-end code.** As mentioned, if you need to use an API key (for Stripe, OpenAI, etc.), **do not paste it into a prompt** – Lovable will warn you if you try ¹⁴⁰ ! Instead, store secrets in Supabase or Lovable’s secure storage. For example, prompt: *“Store the OpenAI API key as a Supabase secret and create an edge function to use it. Do not put the key in the React code.”* The AI will guide you to do exactly that (store secret, call via function) ¹⁴¹ ¹⁴² . This keeps keys out of client-side code which is visible to users.
- **Use Supabase RLS for data security.** When designing your schema, think in terms of which user roles can read or write which data. Supabase’s RLS policies enforce this at the database level. Before deploying, review these policies (AI can help summarizing them if you ask). Lovable’s documentation suggests carefully reviewing RLS policies **before publishing** your app ¹⁴³ . If you’re unsure, ask the AI: *“Check if all Supabase tables have appropriate RLS policies and suggest any missing restrictions.”* It might point out that your “tasks” table allows any logged-in user to read all tasks unless you add a policy to restrict to project members ¹⁴⁴ .
- **Run Lovable’s Security Scan.** Lovable provides an automatic security scan when you publish, integrating Supabase’s security advisor results ¹⁴⁵ ¹⁴⁶ . Pay attention to those warnings – e.g., it might warn about a storage bucket being public or an RLS policy issue. Resolve **all** warnings before confirming publish ¹⁴⁷ ¹⁴⁸ . The scan also includes Lovable’s own AI-powered review beyond Supabase’s checks ¹⁴⁹ ¹⁵⁰ , which can catch things like missing input sanitization, XSS vulnerabilities, etc. You can manually trigger this review any time by prompting, *“Review my app’s security.”* The AI will then analyze code for common flaws (escaping user input, using HTTPS for API calls, proper authentication flows) ¹⁵¹ ¹⁵² .
- **Follow secure coding guidelines in prompts.** Encourage best practices: *“Use prepared statements or parameterized queries for any database operations.”*, *“Escape any user-provided content when rendering HTML.”* If building auth flows, rely on tested libraries (Supabase Auth or Clerk) rather than custom code, to avoid mistakes. The AI by default uses secure patterns (for example, Supabase JS library which uses JWTs securely), but if you ever see it doing something sketchy (like directly embedding a password in a client-side check), correct it immediately.

Your **system design should consider least privilege** – for instance, not every component needs to fetch all data. If you have admin features, perhaps separate them into an admin section that loads admin data, rather than sending admin-only data to the front-end for regular users. The knowledge file is a good place to note these boundaries, so the AI knows, e.g., *“Investor role can only read their own data”* and will implement checks accordingly ¹³¹ .

Finally, **stay updated on Lovable's security features**. As Lovable evolves, new features might include built-in scanning or protections. The Security page in docs mentions that even with no warnings, you should always do a manual review for production apps ¹⁵³. A good habit is to ask the AI to do a *code security audit* similar to the architecture audit before launch: *"Scan all code for any potential security vulnerabilities, like XSS, SQL injection, or leaking sensitive info."* It might catch subtle issues (for example, forgetting to validate image file types before upload).

4.5 Deployment Best Practices: With design solid and your app optimized and secure, you're ready to deploy. Lovable makes deployment straightforward, but here are some best practices:

- **Use Lovable's Preview Links during development.** At any time, you can generate a **Preview** link which shows the latest build of your app without affecting the main published app ¹⁵⁴ ¹⁵⁵. This is extremely useful for testing with others or on different devices. For instance, after implementing a feature, use preview to share with a colleague or on your phone. It's instant and separate from the official URL.
- **Publishing your app:** When you click **Publish**, Lovable will build and host your app at a URL like `yourproject.lovable.app` ¹⁵⁶ ¹⁵⁷. Follow the steps in the UI: confirm publishing, wait ~a minute for it to complete ¹⁵⁸ ¹⁴⁵. After publishing, you'll get an "Update" button for subsequent deploys ¹⁵⁹. Only publish when you consider the app in a shareable state (there's no cost to publishing, but you generally don't want half-implemented features in production).
- **Check the Security Scan on publish.** As mentioned, during the publish flow Lovable will display any security warnings (especially for Supabase-connected apps) ¹⁴⁵. Don't ignore these. If warnings exist, ideally cancel publishing and fix them, then attempt publish again. You *can* publish despite warnings by confirming, but that's not recommended unless you're just testing.
- **Custom Domains:** For a professional app, you likely want a custom domain. Lovable allows you to connect a custom domain through providers like Netlify or Vercel ¹⁶⁰. The typical process is: publish on Lovable (so code is on GitHub), then **export or integrate that repo with Netlify/Vercel** which handle the domain and CDN. The docs have a guide for custom domains ¹⁶¹. Essentially, you add your domain, configure DNS, and your Lovable app will be accessible at `yourdomain.com`. Lovable's "Custom domain" integration automates some of this via Netlify (you'll provide domain and it sets up deployment) ¹⁶⁰. Make sure to republish (deploy) after renaming your project or setting the domain so that the domain link works ¹⁶².
- **Post-Deploy Monitoring:** After deployment, test the live site thoroughly. Sometimes environment differences (e.g., a Supabase production vs. dev instance) can cause issues. If you have error logging or tracking (Lovable doesn't include it by default, but you could integrate something like Sentry via npm), watch for any errors in production. Alternatively, manually check the browser console on the live site for any errors.
- **Launched & Community Sharing:** Lovable has a *Launched* apps showcase (essentially a built-in Product Hunt style listing) ¹⁶³. Once your app is live, you can submit it to **Lovable Launch** ¹⁶⁴ ¹⁶⁵ to gain visibility from the Lovable community. This can drive early users and feedback. It's also a way to allow others to **Remix your project** if you make it public – which is a compliment and a way to contribute back (someone might remix and improve something, or use it as a learning example) ¹⁶⁶.

- **Centralized Plan for Teams:** If working in a team workspace, all members share a credit pool and can deploy. Decide on a deployment strategy: maybe only Owners or Admins trigger publishes (to avoid overwriting each other) ¹⁶⁷ ¹⁶⁸. Use the pin feature to mark a version as “Release Candidate” before publishing, so you can roll back if needed ¹¹⁵. On Teams plan, multiple editors can publish, so communication is key – perhaps establish that you only deploy from the `main` branch and use branches for features (Lovable Labs supports branch switching with GitHub) ¹⁶⁹ ¹⁷⁰.

4.6 Continuous Integration & External Deploys: Some teams use Lovable for development but deploy externally for more control (for instance, deploying on Vercel with their own CI pipeline). Thanks to the **GitHub Integration**, every change in Lovable is pushed to a Git repo ¹⁷¹ ¹⁷². You can set up CI/CD (GitHub Actions, etc.) to run tests and deploy to your own servers or services whenever Lovable pushes. This is advanced, but mentionable: Lovable doesn’t lock you in to their hosting. It’s flexible – you can host the app anywhere once you have the code ¹⁰. For initial stages, Lovable’s hosting is simplest, but as you scale, you might integrate with a more elaborate devops setup.

4.7 Mobile Deployment (PWA & App Stores): If you intend your Lovable app to be used like a mobile app, note that Lovable apps are web apps. However, they can be made into Progressive Web Apps (PWAs) which users can “Add to Home Screen” on mobile. Lovable doesn’t automatically generate a manifest or service worker for PWA by default, but you can prompt the AI to: “Add PWA support: generate a `manifest.json` and a service worker for offline caching of assets.” This is optional but nice for mobile usability.

For publishing to **Apple App Store or Google Play**, since Lovable outputs web apps, you’d use a WebView wrapper (like Capacitor or Cordova). The Lovable docs describe a workaround: you essentially finalize your web app, then use Xcode and Android Studio to wrap it in a native WebView container ¹⁷³. Key steps include having an App Store developer account, converting the web app to a native app (with an embedded webview and proper config), then submitting to stores ¹⁷⁴ ¹⁷⁵. It’s not “one-click” but it’s doable. The guide suggests thorough testing of responsiveness and performance beforehand ¹⁷⁶, and highlights Apple’s stricter process vs. Google’s faster approval ¹⁷⁷. If mobile distribution is important, plan time for this wrapping and review process. (And perhaps use a prompt to the AI: “What should I keep in mind to publish this as a hybrid app on iOS/Android?” It might remind you to use a responsive layout, avoid unsupported features, etc.)

4.8 Optimizing After Launch: Once your app is live and users join, use analytics to identify slow or problematic areas. You can instrument things like Google Analytics or custom tracking (the AI can integrate simple analytics script if asked). For example, track how long API responses take, or memory usage if applicable. **SEO** (Search Engine Optimization) might become important if your app is web-facing – Lovable can help generate meta tags and sitemap as discussed in Chapter 5’s SEO section. Just include in prompts: “Follow SEO best practices for the landing page” (Lovable will add meta title/description automatically) ¹⁷⁸. You can also ask it to generate a `sitemap.xml` and `robots.txt` once your site structure is stable ¹⁷⁹ ¹⁸⁰.

In summary, good system design in Lovable comes from guiding the AI to implement **structured, maintainable architecture**, continuously **reviewing** that architecture (with the AI’s help or manually), addressing **performance** and **security** concerns as first-class citizens, and having a smooth **deployment** strategy. With these in place, your AI-built project won’t just work – it will be scalable, secure, and easy to evolve. Now that we have a solid foundation, let’s explore how to connect your Lovable app with other tools and services (integrations) in the next chapter.

5. Integrations: GitHub, Supabase, Stripe, Clerk, etc.

One of Lovable's strengths is its ability to integrate with external services and APIs to extend your app's functionality. From built-in integrations (like GitHub for version control or Supabase for your database) to third-party services (payments, auth systems, AI APIs), Lovable provides patterns to include these without heavy lifting. This chapter will discuss common integrations and best practices for using them in your Lovable projects.

5.1 GitHub Integration (Version Control & Collaboration): Every Lovable project is backed by a Git repository behind the scenes. By linking this to your own GitHub account, you get **full version control, backup, and collaboration capabilities** ⁹ ¹⁷¹. When you integrate GitHub, Lovable will automatically push every AI edit as a commit to the repository, and pull any changes from GitHub into Lovable in real-time ¹⁸¹. This means:

- You have a complete **history of changes** and can roll back to any previous commit if needed (either in GitHub or via Lovable's version history UI) ¹⁸² ¹⁸³. This is a lifesaver if an AI edit breaks something; you can diff the changes on GitHub and revert the commit that introduced the bug.
- Multiple developers can work on the code using normal Git workflows. For example, someone could create a feature branch on GitHub, code some changes manually, merge to main – and Lovable will sync those changes into the project editor automatically ¹⁸⁴. (Just be cautious to commit changes in Lovable before pulling new ones, to avoid conflicts.)
- You can run CI/CD or deploy outside Lovable using the repo. Some teams treat Lovable as a code generator and use the GitHub repo to integrate with their existing pipeline (tests, code reviews, deployments on their own infra).
- You're **not locked in** – if you ever decide to stop using Lovable's AI, you still have the entire codebase in a GitHub repo and can continue development normally ¹⁸⁵. This flexibility is great for risk management.

How to integrate GitHub: In Lovable, go to the project settings and find GitHub integration. You'll be prompted to **connect your GitHub account** and authorize the Lovable GitHub App ¹⁸⁶ ¹⁸⁷. You can choose to allow access to all repos or select one. Lovable will then create a new repo for your project (you choose the name) and do an initial push of the code ¹⁸⁸. After that, every change syncs. It's essentially real-time: edit in Lovable -> commit appears on GitHub; push commit to GitHub -> it shows up in Lovable in seconds ¹⁸⁹.

Tips for GitHub integration:

- Commit messages for AI edits are autogenerated (often just "AI edit" or listing the files). Consider manually editing commit messages in GitHub for clarity, or squashing commits when merging branches, since the AI may produce many small commits.
- Use **branching carefully**. Lovable primarily tracks one branch (usually `main`). In Labs, there is a feature to switch branches within Lovable ¹⁶⁹ ¹⁷⁰, but it's marked experimental. If you use it, commit all changes before switching, and be aware that branch switching replaces the project files with the other branch's content. Many teams simply do AI development on `main` (since it's quick anyway) and use branches only for manual edits or reviews outside Lovable.
- Protect your main branch on GitHub if you want to review AI changes – you could require pull requests instead of allowing direct commits. In that case, you might not give Lovable write access to

main, but rather to a dev branch. Then you merge PRs into main after code review. This is more advanced but possible. Lovable's default is to commit directly.

- Pin stable versions in Lovable after major milestones (Lovable UI allows “pinning” a version which corresponds to a git tag) ¹¹⁵. This, plus GitHub releases, can mark important releases of your app.

Overall, GitHub integration **brings transparency and safety** to AI coding – you can always see what code was generated and collaborate like any software project ¹⁸² ¹⁹⁰. We highly recommend enabling it, even if you're solo, for the backup alone.

5.2 Supabase Integration (Database & Auth): Supabase is to Lovable what a traditional backend would be to a front-end – it provides a **PostgreSQL database, authentication system, storage, and serverless functions (Edge Functions)**. Lovable has a native integration with Supabase that makes it almost plug-and-play to get a full-stack app working ⁸⁶ ⁸⁷:

- **Database:** When connected, you can prompt Lovable to create tables and the AI will use Supabase's JS client to perform queries (e.g., inserting, reading). The beauty is Lovable can generate the SQL schema migrations for you based on your prompts ⁸⁷. For example, “Set up a `projects` table with columns: `id (uuid)`, `name (text)`, `owner_id (references users)`.” The AI will create that in Supabase, and you'll see it in the Supabase dashboard. Your Lovable app can then read/write to it with the provided API (no custom backend code needed unless you want to enforce something via Edge Functions).
- **Authentication:** Supabase provides out-of-the-box email/password auth (and other providers). Lovable can scaffold auth flows easily. If you prompt “Add signup and login using Supabase Auth”, the AI will add the UI components (sign-up form, login form) and use Supabase's auth library to handle it ¹⁹¹ ¹⁹². You can also leverage third-party auth via Supabase (OAuth with Google, etc.) by some configuration. Clerk (discussed later) is an alternative for auth with more UI, but Supabase Auth works well for many cases. The integration ensures that when a user logs in, the JWT token is managed and Lovable's serverless functions can authenticate requests.
- **Storage:** Supabase Storage is for files (images, etc.). You might prompt, “Allow users to upload a profile picture using Supabase Storage.” The AI will include the Supabase storage logic (upload file to a bucket, get a public URL or require logged-in user to retrieve it depending on your bucket settings) ¹⁹³. Security note: manage your storage bucket rules (e.g., public vs private). The AI will default to what Supabase's JS docs suggest, but double-check RLS for storage or use Supabase's policies on buckets.

Connecting Supabase: In Lovable, click **Connect Supabase** in the Integrations menu. You'll log in to Supabase and either select an existing project or create a new one on the spot ¹²⁴ ¹⁹⁴. Once linked, Lovable stores the Supabase project reference and keys securely – the AI will not show or hardcode the service key, it uses an environment variable or context that Lovable manages ¹⁴⁰ ¹³. After a few seconds, your Lovable project is “Supabase-enabled.” You can verify by checking that you have a `supabase` object or import available in your code.

Using Supabase in prompts: After integration, you can instruct the AI at a high level and it will handle the under-the-hood details:

- “Create a notes table in Supabase with fields: `id (uuid)`, `user_id`, `content`, `created_at`. Then allow users to Create, Read, Update, Delete notes from the UI.” – The AI will create the table (via an Edge Function or the Supabase JS client with an `createTable` call), set up a form and list in the UI, and ensure that the `user_id` is the current logged-in user's ID for each note (especially if you mention it). It might

even add RLS policies automatically (e.g., only owner can access their notes) – if not, you should add those.

- *“Implement real-time updates for the notes list using Supabase.”* – Supabase can stream live changes (with `on(*)` subscription). The AI can add a real-time listener so that if another user adds a note, it appears without refresh ¹⁹⁵. This is one of Supabase’s cool features, and Lovable’s AI is aware of it if prompted for “real-time”.
- *“Add a Supabase Edge Function to send a welcome email on user signup.”* – This is more advanced but shows integration: Supabase Edge Functions are serverless functions (written in TS/JS) you can deploy to Supabase. Lovable can help draft one. It would involve writing a file in the `supabase/functions` directory of the repo. The AI might not do this by default unless specifically asked (since Edge Functions require CLI to deploy), but it might simulate it by using a serverless function in Lovable’s environment (Lovable might allow you to create “server functions” that run via Supabase).

Why Supabase with Lovable: It **eliminates boilerplate**. You don’t have to set up an Express server or an ORM – you just describe your data needs, and Lovable handles linking the front-end to the database. Supabase’s web dashboard lets you inspect and manage the data easily as well ¹⁹⁶ ¹⁹⁷. In essence, Lovable + Supabase gives you a full-stack platform: React on the front, Postgres on the back, all driven by prompts.

Common pitfalls & best practices:

- **Supabase URL and Key:** Lovable will prompt you to log in and auto-configure these. Don’t hardcode keys. Use the **Add API Key** form in chat if the AI ever asks for a key (for example, when integrating Stripe, it explicitly says to use the in-chat form for the secret key) ¹⁹⁸.
- **Apply DB migrations:** When AI creates or alters a table, those changes occur on Supabase instantly (in the linked project). If you revert code to an earlier version but the DB schema stayed changed, you can get mismatches (like code expects a column that no longer exists). Best practice: after heavy schema prompts, if you use version control rollback, also manually rollback the DB if needed. Lovable’s best practice doc warns that Supabase schema changes are not auto-reverted with code revert ⁸⁸ ¹⁹⁹.
- **RLS (Row-Level Security):** After designing tables, decide on RLS policies. Supabase requires you to enable RLS per table. Lovable might or might not auto-enable and configure it based on context. Check the Supabase dashboard or ask AI: *“Ensure RLS is enabled on all tables and that each table has appropriate policies (e.g., user can only see their data).”* ¹⁴³ ¹⁴⁷ The AI may instruct you to create some policies or even generate policy SQL if you ask. Always test as a non-admin user to verify you can’t access others’ data.
- **Edge Functions vs. Direct Calls:** For most things, the AI will directly call Supabase (like `supabase.from('table').insert(...)`). This is fine for many use cases. But if you need to do sensitive operations (like calling an external API with a secret), it’s better to route it through an Edge Function on Supabase so the secret stays server-side ¹³. The Lovable security scan will warn if it sees something like a secret used in front-end code. So design accordingly: e.g., for Stripe webhooks or OpenAI calls, use Edge Functions integrated via Supabase (or Lovable’s serverless if you have one).
- **Testing Supabase locally:** There’s an option to run Supabase locally via the CLI for development. But in Lovable’s context, you typically use the hosted Supabase project directly for dev. Just be mindful of quotas on the free tier. If you need to reset data, you can always drop tables in the Supabase dashboard or ask AI to make a quick admin page to delete test data.

5.3 Stripe & Payments Integration: Lovable can set up payment flows via **Stripe** with remarkable ease. When integrating Stripe, the goal might be to accept payments (one-time or subscriptions) from your users. The **Stripe & Payments integration** in docs shows that Lovable can **auto-generate checkout and billing logic via chat** ²⁰⁰ ²⁰¹ .

Before you start: Get a Stripe account and find your **Stripe Secret Key**. **Important:** *Do not paste the secret key directly into a prompt!* Lovable has a special “Add API Key” input in the chat UI for things like Stripe keys ²⁰² . Use that when the AI asks for it, so it's stored securely (the AI knows to fetch it from secure storage, and it won't be exposed in your code).

Process to set up Stripe in Lovable:

1. **Connect Supabase first.** The Stripe integration relies on Supabase Edge Functions to handle webhooks or secure operations, and storing customer data in your database. The docs note the project must be connected to Supabase before doing Stripe ²⁰³ . If not, do that (as covered above).
2. **Save your Stripe Secret via Add API Key.** In chat, you can type something like “*I want to integrate Stripe for payments.*” The AI will likely respond with instructions or ask for your Stripe secret. At that point, a UI form will appear to input the key (sk_test_xxx...). Once submitted, Lovable warns not to hardcode it and uses it behind the scenes ²⁰² .
3. **Describe what you need:** Lovable supports both **one-time charges (Payment Links/Checkout)** and **subscriptions (Stripe Checkout + customer portal)**. For example:
 4. “*Add three subscription tiers: Free, Pro (\$10/mo), Team (\$30/mo for up to 5 users). Users on Pro or Team have access to premium features.*”
 5. Or “*Create a one-time purchase option for an e-book at \$29. After payment, allow the user to download the PDF.*”

In your prompt, detail the pricing and what should happen post-payment. The AI will then: - Create a **Supabase Edge Function** (in TS) for Stripe checkout using Stripe's API (it might use Stripe Checkout sessions under the hood, or direct charges, depending on how it's prompted) ²⁰⁴ ²⁰⁵ . - Set up any needed Supabase tables, e.g., a `subscriptions` table or add a `role` field to your users to mark who is Pro. - Add UI components: maybe a “Upgrade” button that opens Stripe Checkout, and a webhook handler to update the user's status on payment success. - It might rely on polling instead of webhooks by default (the docs say webhooks are opt-in; Lovable may choose to poll the subscription status via Edge Function periodically) ²⁰⁶ . - It will certainly advise to **test in Stripe's test mode** and provide guidance (like using the test card 4242 4242 4242 4242) ²⁰⁷ .

For example, Lovable might generate an Edge Function called `createCheckout` that takes a price ID, creates a Stripe Checkout session, and returns the URL. The front-end will call this function (Lovable knows how to call Supabase functions) and then redirect the user to the Stripe Checkout URL. On return, another function or a client query might verify the payment and update Supabase (like set `user.isPaid=true` or insert into a `subscriptions` table with `status='active'`). The docs note Lovable generates the necessary DB tables and RLS for linking Stripe customers to users ²⁰¹ .

1. **Follow the AI's prompts for configuration:** It might output steps like:

2. "Add your Stripe secret key via Add API Key" (we did).
3. "Make sure you have created the product and price in Stripe Dashboard" – sometimes you need to pre-create the price IDs unless using Stripe's API to create them on the fly. The AI might assume you have price IDs ready or ask for them. You can get them from Stripe dashboard (or instruct the AI to use Stripe's API to create products on the first run).
4. It may instruct you to deploy the Edge Function (Lovable might deploy it to Supabase automatically when you publish the project).
5. **Testing:** Always test in **Stripe Test Mode**. Lovable emphasizes using test mode and the test card (4242...) ²⁰⁷. You'll need to publish or use a preview link for the Stripe Checkout (because Stripe needs a reachable URL). The integration won't work on a local-only environment. Preview links are fine since they are internet accessible.

Stripe integration example (subscription): Suppose we prompt for subscriptions. The AI will: - Add a "Pricing" section in your app UI with the tiers. - For each tier's "Upgrade" button, integrate with an Edge Function that creates a Stripe Checkout session for that tier's price. - It might also set up a **Stripe Customer** mapping: linking the Supabase `users` table row to a Stripe Customer ID. This is often done by sending the `user.id` (or email) to Stripe when creating the checkout, so Stripe knows who to tie the sub to. The AI might store the returned Stripe customer ID in the user's record. - Implement after-payment flow: Likely a webhook or a polling to update the user's role/tier in Supabase. The docs hint at two approaches: default is polling via Edge Function without webhooks ²⁰⁶, but you can opt into webhooks. For simplicity, the AI might use polling (meaning, when user returns from Stripe Checkout, maybe it calls Stripe's API to confirm the session payment status via the Edge Function). - Provide a **Customer Portal** link for users to manage billing (Stripe has a customer portal that can be integrated so users update card or cancel). The AI might create a "Manage Subscription" button that uses Stripe's Billing Portal.

Key takeaways and tips from Lovable's Stripe integration doc:

- Use the **chat-driven flow** – indeed, just describe what you want in plain English, Lovable does the heavy lifting of code and database changes ²⁰⁰.
- **Do NOT paste secret keys in chat** – use the provided secure input ²⁰².
- **Webhooks are opt-in** – meaning Lovable's default solution may not automatically use webhook endpoints unless requested. They might poll subscription status using scheduled Edge Function calls or check on login, etc. If you need instant webhook handling (like for sending a welcome email on payment), you might have to implement that. Lovable recommends using Supabase Edge Functions for secure handling of Stripe events (which presumably the AI can set up if asked) ²⁰⁸.
- **Debugging:** Use browser console network logs, Supabase Edge Function logs, and Stripe Dashboard logs to troubleshoot if something doesn't work initially ²⁰⁶. The AI might not get everything perfect first time, especially if your Stripe setup (products/prices) differ from its assumptions. Be ready to adjust price IDs or endpoints.

Finally, **test thoroughly** using Stripe's test mode. Once it works, deploy to production (Stripe live mode) by swapping test keys for live keys – likely in the Lovable account settings or by updating the API key variable to the live secret and adjusting any test environment flags.

5.4 Authentication & User Management Integrations – Clerk: While Supabase provides basic auth, **Clerk** is an integration specializing in authentication and user management with rich pre-built UI. If you want social logins, multi-factor auth, user profile management out of the box, Clerk is a great choice ²⁰⁹.

Setting up Clerk: You create an application in Clerk (via clerk.com) and get a **Public API Key** (for frontend) and a secret (for backend). The process from the docs:

- In Clerk's dashboard, you configure your app (enable OAuth providers like Google, enable "Organizations" if you want team accounts, etc.) ²¹⁰ ²¹¹.
- Copy the **Public API Key** – this is used on the front-end to initialize Clerk.

Using Clerk in Lovable: You'll tell Lovable's AI something like: *"Integrate Clerk for authentication and user management. Use Clerk's prebuilt components for sign-in/sign-up, and enable organization (team) support."* The AI will:

- Prompt you (or instruct) to add the Clerk Frontend API Key, likely via an Add API Key form or a secure input (similar to Stripe's). You provide the public key there (public keys are not super sensitive, but still).
- Include Clerk's SDK in the project (Clerk provides React components and hooks). Possibly the AI will add an `<SignIn />` component or use Clerk's `<SignUp />`, etc., and wrap your app in a `<ClerkProvider>` with your API key.
- It can replace Supabase Auth entirely – if you were using Supabase Auth, you might switch to Clerk to handle auth (Clerk can integrate with Supabase as well to create a user row on sign-up via webhooks, but that might be more advanced; often you might use Clerk's own user database separate from Supabase's).
- Add UI routes for Clerk's components (like `/sign-in`, `/sign-up`, maybe a `/user-profile` page with `<UserProfile />` component).
- If you mention **"Waitlist mode"**, the AI will know from docs that Clerk can restrict sign-ups and require admin approval (Clerk's "waitlist" feature) ²¹² ²¹³. It might configure that (you'd have toggled it in Clerk settings; the AI might just note it).
- If you mention **"organizations"** (teams), the AI will enable Clerk's organization feature (which allows users to create organizations and invite others) ²¹⁴ ²¹⁵. The integration would include prompts to Clerk's UI components like `<OrganizationSwitcher />` or guidelines to ask the AI for how to prompt for joining/creating orgs after sign-up ²¹⁶. The AI can scaffold a flow: after sign-up, if using orgs, ask user to create or join an organization (Clerk provides components for that).
- The AI will **scaffold role-based UI** if you tell it what roles or what features to restrict. E.g., *"Only admins can access the admin dashboard, and Clerk has roles Admin and Member in orgs."* It can enforce that via Clerk's session info (Clerk's JWT can have roles, or you query Clerk's user object).

Clerk integration benefits: - Beautiful pre-built components (fully themed forms etc., saving design time). - Handles tricky auth features (password resets, email verification, social OAuth, multi-factor) which you'd otherwise have to custom-build. Clerk's free tier is generous (and they handle the email sending, etc.). - Clerk can integrate with other services: e.g., if using Stripe, Clerk soon or already can link to Stripe (the docs mention integration with Stripe coming soon ²¹⁷). - Clerk offers **user profiles** and **impersonation**, etc., which can be useful in admin tools.

Tips with Clerk: - Follow Clerk's quickstart: the AI usually does, by adding `<ClerkProvider frontendApi="...">` around your app (maybe in `_app.jsx` if using Next, but in Lovable's case likely in

an index file). - The Public API Key (frontendApi) should be used in the provider, and the Secret API Key can be used in Edge Functions if needed (for example, to verify a Clerk session server-side). If you need the secret, store it securely (similar to other keys). - Test the sign-up and login flows thoroughly. Ensure your site URL is configured in Clerk (Clerk needs to know what domains to allow for auth callbacks). - If enabling **“Waitlist mode”**, test that sign-ups are indeed marked pending and that you as admin can approve via Clerk Dashboard (the AI mentioned steps to configure waitlist in Clerk settings) ²¹² ²¹⁸ . - If using **“Organizations”**, check that creation, invitation, and role assignment works. The AI steps for organizations (enable in dashboard, then in app code prompt user to create org, join invites, etc.) will get you started ²¹⁹ ²²⁰ . Clerk’s UI handles most of it, but you might need to write logic to associate application data with org IDs.

5.5 Other Integrations: Email (Resend), Workflows (Make), AI APIs (OpenAI/Anthropic), etc:

- **Resend (Transactional Email):** Resend is an email sending service. Lovable can integrate Resend for sending emails from your app (e.g., confirmations, newsletters) ²²¹ ²²² . To use it:
 - Get a Resend API Key, add via secure input.
 - The AI will likely create a Supabase Edge Function to send an email via Resend’s API (so your key stays secret). For instance, in the CRM example from docs, they used Resend to send a confirmation email on lead submission ²²³ ²²⁴ .
 - The AI might suggest using **React Email** (a library by Resend to create email templates in React) ²²⁵ . If you prompt for a nicer email design, it might use React Email components to format the email HTML.
 - Test by actually submitting the form and checking if the email is received (Resend’s dashboard shows logs).
 - Ensure DNS setup for Resend (verify your sending domain) as the docs mention updating DNS for deliverability ²²⁶ .
 - Resend integration is straightforward: you basically call Resend’s send email endpoint with payload (to, subject, body). The AI will do that likely in an Edge Function or directly via fetch from the front-end if non-sensitive (but API key should remain secret – likely in the Edge Function) ²²⁷ .
- Tip: If using Resend for bulk or ongoing emails, consider using Supabase Edge Functions or `cron` for scheduling, which Lovable can set up in Supabase (Supabase has a Scheduled Functions extension, the AI might not automate that unless asked).
- **Make (No-code Workflows):** Make (formerly Integromat) allows connecting your app via webhooks to 3rd party services in workflows ²²⁸ ²²⁹ . Lovable’s integration with Make is via **webhooks and API calls**:
 - You might have Lovable send an HTTP request to a Make webhook when an event happens (like a deal moved to a new stage triggers an email via Make).
 - Conversely, you might call Lovable’s Supabase Edge Function from Make to do something in your app after an external trigger.
 - The docs provide an example of building a CRM where moving a deal card triggers an automation (send email, AI call with VAPI, etc.) ²³⁰ ²³¹ . They likely did this by having an onDragEnd event in the front-end that calls a Make webhook URL with deal info, and Make handles the rest (like sending an email via Gmail integration).

- To integrate, you'll give the Make webhook URLs or use Make's API. The AI can help: *"On deal update, make an HTTP POST to this Make webhook with the deal data."* It will implement that in the front-end or Edge Function.
- **Security:** If Make workflows call back into Lovable, ensure any incoming webhooks have a secret or your Edge Function verifies the origin, to avoid abuse.
- Using Make is powerful when you want to connect to hundreds of apps (Slack notifications, Notion updates, etc.) without writing custom API calls for each.
- **OpenAI / AI Model Integration:** While Lovable's builder is powered by AI, you can also integrate AI into *your app's functionality* (like building a GPT-powered feature in your app). Lovable's *Prompts & Integrations* doc lists supported AI models – OpenAI, Anthropic, Cohere, etc ²³² ²³³ . For instance:
 - If you want to add an **AI chatbot in your app**, you might prompt: *"Add a feature where users can ask a question and get an answer from GPT-4."* The AI would integrate OpenAI's API by:
 - Asking for your OpenAI API key (again, via Add API Key form).
 - Creating an Edge Function (to keep the key secret) that calls OpenAI's chat completion API with the user's question ²³⁴ ²³⁵ .
 - On the front-end, a simple form to send the question to that function and display the response.
 - If you have a specific model or provider (Anthropic's Claude, for instance), Lovable supports those too – maybe by letting you select or by specifying in the prompt e.g., *"Use Anthropic's Claude model for summarization"*. The integrative doc shows patterns like `[AI model]` placeholders for various tasks ²³⁶ ²³⁷ .
 - It also mentions specialized tools like **Groq** (an AI service focusing on speed using Llama models) or **Deepseek, Mistral**, etc., which presumably Lovable has knowledge of usage patterns for ²³³ ²³⁸ .
 - Key point: Always use secure methods to call AI APIs – no keys in client code, use serverless functions or at least Supabase secure environment. Lovable's integration doc specifically highlights *"Use function calling if API returns structured data"* and other advanced tips ²³⁴ ²³⁵ – which implies it encourages using OpenAI's function calling feature where appropriate, and doing calls in Edge Functions with keys stored in Supabase secrets ²³⁹ .
 - The doc snippet also shows example prompt usage: e.g., *"Use GPT-4 and function calling to extract nutrition info from a meal description and save results to backend"* ²³⁵ – the AI would craft a function call to get structured output and then store it.
- **21st.dev (UI components library):** 21st.dev is a library of Tailwind UI components. Lovable's integration with it allows you to quickly use high-quality prebuilt sections (hero, nav, etc) by just prompting. From the integration doc ²⁴⁰ :
 - You might say *"Use a 21st.dev component for the pricing section with hover effects."* The AI will likely know a prompt from 21st.dev's library (21st.dev might have specific prompt templates to generate components).
 - Essentially, it's like copying beautiful Tailwind component code from a library without searching for it manually – you ask the AI and it inserts or adapts it ²⁴¹ .
 - The earlier "Landing Page blueprint" in Chapter 7 references using 21st.dev to polish design (hero, testimonials, etc.) ²⁴⁰ ²⁴² . The AI can incorporate those by prompt or by reading their docs which Lovable might have.

- Tip: You might find 21st.dev components at their site and literally provide the component name to AI: “Add a 21st.dev hero section component with a background image and a CTA.” The AI will include appropriate HTML/JSS and Tailwind classes.
- This integration saves time on UI/UX polish and ensures consistency, since these components are expertly designed.
- **Chrome Extensions (Lovable Add-ons):** Not an app integration, but developer tooling. As seen in the “Chrome Adds-on” doc ²⁴³ ²⁴⁴, there are community-made extensions to boost Lovable development. For example:
 - **Lovify by Talisha:** Adds features like “Spark Prompts” (suggest prompt completions), voice-controlled debugging (“Hey Lovify, fix this bug...” which likely uses speech-to-text to prompt Lovable) ²⁴⁵, and a nifty *Import from GitHub* function (to bring an existing project into Lovable quickly) ²⁴⁶.
 - **Lovable.dev Add-ons by Rezaul:** Provides voice input, advanced chat search, color picker, SEO meta generator ²⁴⁷ ²⁴⁸.
 - **Lovable Prompts & Lovable Helper:** For managing prompt templates and quick actions (Ctrl+Enter to run prompt, etc.) ²⁴⁹ ²⁵⁰.
 - **Prompt2MVP:** Helps capture screenshots and turn them into prompts, plus choosing tech stacks, etc ²⁵¹.
- These aren’t integrated *into your app*, but into your development workflow. They highlight a growing ecosystem.

In conclusion, Lovable’s integration capabilities turn it into a “center of a hub” connecting various services. The approach is usually: **obtain API keys -> provide to Lovable securely -> prompt AI to use the service’s API or SDK**. The AI’s familiarity with these patterns (as evidenced by the documentation citations) means you often don’t have to write glue code from scratch – just describe what you want to happen.

By leveraging integrations, you can add production-ready functionality in minutes: payments with Stripe, rich auth with Clerk, content emails with Resend, automation with Make, and intelligent features with AI APIs. Always keep security in mind (no secret in front-end, limited scopes, etc.), and test each integration thoroughly in its sandbox (most services have test modes or dev environments). As Lovable’s motto suggests, these integrations let “ordinary visionaries achieve breakthroughs” by doing in a prompt what used to require days of coding ²⁵².

Next, we’ll discuss best practices for Lovable projects in general – bringing together everything from planning, prompting, debugging, to community tips.

6. Developer Best Practices for Lovable Projects

To get the most out of Lovable, it helps to follow certain best practices that experienced users have learned. These practices ensure you work efficiently with the AI, maintain code quality, and avoid common pitfalls. We’ll cover tips on project organization, collaboration, coding habits, and leveraging the Lovable community.

6.1 Set Your Foundation: Use the Knowledge File. At the start of a project (and continuously as it evolves), maintain a **Knowledge File** that captures your project's essential context ⁵³. This file is like the brain of your app – Lovable sends it with every prompt to give the AI situational awareness ²⁵³. Include in it:

- **Product vision & description:** What the app does, who it's for. E.g., "This is a task tracking app for remote teams to manage projects."
- **Key features and functionality:** List major features, maybe as user stories. *"User can create a project, add tasks to it, assign tasks to members..."* ²⁵⁴.
- **User roles or personas:** If you have roles (Admin, Investor, etc.), define them and their permissions here ¹³¹.
- **Design guidelines:** Preferred style, color scheme, any specific UI framework (Tailwind, Chakra). *"Use Tailwind CSS with a minimalist aesthetic; primary color is #3498db."*
- **Important constraints:** e.g., "App must support mobile view," "Data must be stored securely with RLS," etc.
- **Project-specific terms or algorithms:** If your app has domain-specific logic (like "In our app, a 'Deal' goes through stages X, Y, Z"), document that.

By feeding this into the knowledge base, the AI is less likely to go off-track. For example, if it knows from knowledge that *"Investors should not see confidential metrics,"* it will incorporate that rule in relevant prompts without you repeating it each time ¹³¹. Make it a habit to update the Knowledge file as you implement new features or rules, so it stays in sync with your app ¹¹⁹. Some builders even ask Lovable's Chat Mode: *"Generate a knowledge file for my project based on what we built so far."* This can produce a nice summary (review it for accuracy).

6.2 Prompt with Purpose (Best Practices for Prompting): This recap of earlier strategies is important enough to list as a checklist ⁵⁴ ³⁵:

- **Be specific:** Always mention exact pages, components, or files you are referring to in your prompt ⁵⁴ ²⁵⁵. Instead of "fix the page," say "In `/dashboard` page, fix the alignment of the stats cards." Precise references eliminate ambiguity.
- **Use natural language, but with clear directives:** It's fine to phrase things conversationally (the AI is quite adept at parsing natural language), but make sure the request itself is unambiguous. For example, *"I want only users with role Investor to access the `/investor-dashboard`. Prevent Admins from seeing it."* is clear about the expected behavior ¹³¹.
- **Include screenshots or images when describing UI issues:** A picture is worth a thousand words in debugging CSS or layout. Lovable allows image uploads – use them. *"See this screenshot: the sidebar is missing an icon. Please add it back."* The AI will understand the visual context better ⁸¹ ¹⁰⁴.
- **Add guardrails in prompts:** If you know some file shouldn't be touched, or a piece of logic must remain unchanged, say so. *"Do not edit `shared/Layout.jsx` during this change."* ³⁵. Or *"Keep existing login logic intact while adding the new SSO."* This helps avoid unintended side effects.
- **Repeat critical instructions across prompts:** The AI's context window isn't infinite; if something is really important (like "don't use any external UI library except Tailwind"), mention it again in later prompts if you suspect the AI might deviate ²⁵⁶. Redundancy can be good – better to over-communicate requirements than under.
- **Break tasks into smaller chunks:** As discussed, avoid giant multi-part prompts that do everything at once ⁷³. The best practice is iterative development: implement one feature, test it, then move to the next. Use Chat Mode in between to verify that the plan is sound before executing. A nice pattern

from the docs is the “Feature Breakdown Template” – e.g., **“Feature: Add user profile pages.**

Breakdown: 1. Create `ProfilePage` component. 2. Show user info (name, email). 3. Allow editing of bio. 4. Save changes to database. 5. Test with a demo user.” ²⁵⁷ . You or the AI can create such checklists and then tackle them one by one.

- **Consider roles and state in prompts:** If your app has multiple user types or states, specify which one the prompt concerns. *“As an Investor user, the dashboard should show portfolio value, but not editable settings.”* By clarifying the context, the AI will produce code conditioned on user role (e.g., render different components based on role) ¹³¹ . Similarly, if something should happen only when X state is true, mention it.

6.3 Use Chat Mode Early and Often (Thought Partner): Don’t reserve Chat Mode for when things break – actively use it while building to **think through designs and logic** with the AI ²⁵⁸ . For example:

- **Brainstorm approaches:** *“What’s the best way to implement a real-time chat feature here?”* The AI might discuss using Supabase’s real-time or a service like Pusher, giving pros/cons ²⁵⁹ .
- **Design DB schema:** *“Help me design the database schema for a multi-tenant app where each org has many users.”* The AI can propose tables and relationships, which you can then implement (or even have AI implement).
- **Debug step-by-step:** As covered in Chapter 3, Chat Mode is great after a couple failed fixes. E.g., *“We’ve tried X and Y and still have the bug. Let’s analyze it.”* ²⁶⁰ The AI can recap what’s been done and suggest fresh ideas.
- **Validate ideas before committing:** If you’re about to prompt something complex, you can first outline it in Chat: *“I’m thinking to add feature X by doing A, B, C. Does that sound right?”* The AI might refine your plan (like a rubber-duck debugging session) ²⁵⁹ ²⁶¹ .
- **Use Chat Mode to avoid consuming edit credits unnecessarily:** This is a big one – maybe 70% of your time can be in Chat discussing, and only 30% executing changes ²⁶² . Because if you blindly try editing and it fails twice, that’s wasted credits and time. Instead, talk it out: *“The login function still returns undefined. What could I try next without changing code immediately?”* ²⁶² .

One workflow tip from docs: If you often forget to use Chat Mode, adopt a format in edit mode prompts: *“Plan: ... Don’t execute yet.”* or use a special phrase like *“Investigate but don’t write code yet”* ²⁶³ . This triggers the AI to give an analysis in Chat style, even if you’re in edit, effectively preventing it from making code changes until you’re ready ²⁶⁴ . You can also explicitly instruct: *“Before writing any code, explain your plan.”* – a great way to get the AI to dump its reasoning which you can sanity check.

6.4 Avoid Infinite Fix Loops – Know When to Reset: If you find yourself in a cycle of the AI making a change -> app still broken -> AI tries another minor tweak -> still broken (the “infinite error loop”), step back. The best practice is:

- **Switch to Chat Mode** after 2–3 failed attempts ²⁶⁰ ²⁶⁵ .
- Paste the error and ask for a fresh perspective: *“Take a step back: what is the root cause of this error?”* ²⁶⁶ ²⁶⁷ . This often breaks the loop.
- If the AI seems stuck in reasoning too, consider a **remix** or revert as mentioned. Sometimes starting the feature over (with lessons learned) is faster than patching a flawed implementation ⁹⁰ ¹⁰² .
- You can also manually intervene: read the code yourself, add some `console.log` or breakpoints via Dev Mode, find the issue, then explain it to the AI: *“I debugged and found that `userId` is null at the point of saving. Modify the code to fetch the correct `userId` before saving.”* This kind of directive, with

the insight you gained, will likely fix it. The AI is good, but your logical deduction plus AI's coding speed is the real power combo.

6.5 Use Visual Edit for UI Tweaks: Not every change needs a prompt. Lovable's **Visual Editor** lets you click on elements and adjust them. Best practices for Visual Edit:

- Use it for purely presentational changes: text content, colors, margins, flex alignment, etc. It's often faster to just drag or select an option than to craft a prompt for such minor CSS tweaks [62](#) [268](#) .
- Batch small adjustments: You can make multiple edits in one visual editing session then save (this will be one commit). E.g., change a button text, its color, and font size all via the editor.
- Visual Edits are **credit-free** and have undo – meaning you can experiment without penalty [268](#) . If the AI gave you a layout slightly off, instead of another prompt and uncertain result, just fix it visually.
- That said, avoid doing large structural changes via Visual Editor (like reordering major components) because it can get cumbersome. Those are better in code via prompt or manual code mode. But small UI polish – definitely use Visual Edit as your friend.

6.6 Version Control Wisdom (GitHub + Version History): Because Lovable auto-commits every change (if GitHub integrated), you have a rich history. Best practices here:

- **Pin stable milestones:** After completing a major feature or fixing a tough bug, pin that version (Lovable UI has a “Pin” button) [115](#) . This makes it easy to identify “good” states to revert to if needed.
- **Compare diffs for debugging:** If something broke recently, use Lovable's history or GitHub to compare the last working version vs current [269](#) . The AI can even assist: *“Compare the version from yesterday to now and see what changed in login logic.”* (It might not have a built-in diff tool, but you could manually diff and then ask AI about the changes.)
- **Use branches for risky changes:** If you plan to attempt a major refactor via AI, consider doing it on a separate branch (Lovable Labs feature or manually via GitHub) so you don't lose a working main. You can always merge later. But note Lovable's branch switching caveats (commit everything, etc.).
- **Leverage GitHub actions for tests:** If you have any automated tests (you can even prompt AI to write simple tests), set up a GitHub Action to run on push. That way, if an AI edit breaks a test, you'll know immediately. It brings some CI rigor into the AI loop.
- **Sync issues:** If you do manual edits in GitHub and Lovable at same time, be mindful of sync. Ideally, one person or process at a time. If you get merge conflicts (rare if you're careful), you might need to resolve in GitHub. As a rule, commit in Lovable before pulling external changes in.

6.7 When All Else Fails, Remix: Lovable's *Remix* feature is basically “fork this project into a new one, starting at the current state” [90](#) [102](#) . This is extremely useful if: - The project's prompt history is cluttered or the AI seems stuck on a wrong approach. - You want to experiment heavily without messing up the main project. - You need to **disconnect Supabase** and try a fresh schema (remixing doesn't carry over the Supabase link, so you can connect to a new Supabase project or none, giving a clean slate for DB if needed) [102](#) [270](#) .

Using Remix gives you a psychological fresh start too – sometimes leaving the baggage behind helps the AI start anew with a better plan (especially if knowledge file is cleaned up for the new start). As the docs said, building something the second time is often faster and better organized [90](#) [102](#) .

6.8 Stay Patient, Stay Calm: It's easy to get frustrated if the AI doesn't get it right the first time. But patience is key. If a bug is irritating, take a breather, perhaps even tell the AI *“I am frustrated that this isn't working.”* Interestingly, the docs suggest using an “I am frustrated...” prompt to nudge the AI to focus better

²⁷¹ . It's a psychological trick – by stating frustration and the core issue, the AI might re-evaluate systematically (almost like appealing to its empathetic side, even if it's just text). For example: *"I am frustrated that after all these fixes, the login still fails. Let's systematically check every step of the login flow."* The AI may respond with a calmer, more structured analysis than before.

Remember, debugging and perfecting is part of development – even with AI. The final 5% of polish might take 20% of the time, that's normal ²⁷² . Keep prompts precise and inputs methodical, and you'll get there. Use each failure as learning: update the knowledge file with the gotcha (so next time the AI won't repeat it), write down a new prompt template for similar future tasks.

6.9 Use the Docs & Ask for Help: Lovable's documentation (which we've heavily referenced here) is built into the platform – you can highlight an error and use "Ask AI" on docs, etc. Don't hesitate to query: *"Check Lovable docs for how to integrate X"*. The AI can pull answers from the docs (as we've seen with citations) ¹⁰⁸ . This can be faster than hunting the docs manually.

Also, the **Lovable Community** (Discord, forums) is a great resource ¹⁹ ²⁷³ . Often, someone has built something similar or faced a similar bug. Share your issue (with screenshots or code excerpts) – many in the community, including the Lovable team, are helpful. Plus, explaining the problem to others can clarify it for yourself.

If you're on a paid plan with support, use it (Lovable's priority support for paying users might give direct help) ²⁷⁴ ²⁷⁵ . But the community is a goldmine of tips – sometimes folks share prompt snippets or custom components that you can reuse (maybe check the community examples or the "Launched" apps for inspiration).

6.10 Bonus Tips & Tricks:

- **Voice Input for Prompts:** You can use your operating system's dictation or a Chrome extension to speak prompts out. It's surprisingly effective to just say your idea out loud (in natural language) and let the AI do it – often yielding very fluid results ²⁷⁶ . If you're tired of typing or want to articulate a long prompt, voice can speed things up.
- **Custom Prompt Templates:** Keep a library of prompt templates that worked well (Lovable's Prompt Library doc is a good starting point, but you can have your own). E.g., a template for "Explain code", one for "Lock file X and do Y", one for "Full code audit". Reuse them as needed.
- **Utilize `Select Code` + Prompt:** Lovable has a feature where you can select a block of code and then prompt about it specifically. Use this to focus the AI. For instance, select a function and ask "Refactor this function for clarity." The AI will then target just that code ³⁵ ²⁷⁷ .
- **Mobile and SEO Checklists:** Before considering an app "done," run through a quick checklist:
- Test on a mobile viewport (Lovable has a mobile preview toggle) – fix any glaring responsive issues (often just adding appropriate Tailwind classes or using flex-wrap, etc.).
- Generate SEO metadata: use the AI to create a good meta description, title, maybe JSON-LD schema for rich results if applicable ²⁷⁸ ²⁷⁹ .
- Optimize images: ensure you didn't leave huge images uncompressed (AI sometimes uses large Unsplash links – consider using a smaller size or using Pexels integration as Lovable can search images via Pexels API) ¹³⁷ ¹³⁸ .

- **Accessibility:** Ask AI to check for obvious accessibility issues (missing alt text, proper ARIA labels). You can prompt: *"Audit the app for accessibility best practices."* It might suggest adding alt tags, using semantic HTML for screen readers, etc. ²⁸⁰ ²⁸¹ .
- **Continuous Learning:** Each Lovable project you do, you'll gather new best practices. Add them to your personal checklist. For example, after one project you might note "Always structure prompt with Context/Task/Constraints for big features – got better results." In the next project, follow that from the start.

By adhering to these best practices, you can turn Lovable from just a nifty AI tool into a superpower that amplifies your development productivity. It's like working with an extremely knowledgeable, if sometimes overeager, pair programmer. You provide the vision and guidance; the AI provides the speed and breadth of knowledge. Together, you'll build amazing things faster than ever.

7. Use Case Blueprints (SaaS, Landing Page, Prototype, etc.)

Lovable isn't just for one type of project – you can build anything from a quick prototype to a full-fledged SaaS product. In this chapter, we provide "blueprints" or patterns for a few common use cases: **Landing Pages**, **Prototypes/MVPs**, and **SaaS applications**. Each blueprint highlights recommended workflows, feature sets, and tips specific to that scenario, so you can generalize them to similar projects.

7.1 Landing Page Blueprint: Landing pages (for marketing a product or collecting signups) are a perfect fit for Lovable's rapid UI generation. The goal is often a visually appealing, responsive page with sections like hero, features, testimonials, FAQ, and a call-to-action. Here's how to approach it:

- **One-Shot Generation vs. Section-by-Section:** You have two strategies ²⁸² ²⁸³ :
 - **One-Shot:** Describe the entire page in one prompt – e.g., *"Build a landing page for a space tourism agency targeting wealthy individuals. Include a hero section with a space image and catchy headline, a section on zero-gravity flights, a pricing section, and a contact form. Use a modern, sleek style."* ¹¹ ¹² . Lovable will attempt to generate the full page with all sections. This is quick and often gives a decent draft. Make sure your prompt is detailed: specify the audience, key selling points, desired tone (luxurious, adventurous, etc.). The AI will make assumptions for filler text, which you can later refine.
 - **Section-by-Section:** Tackle each part in sequence. Prompt: *"Create a blank hero section with a background space image, title 'Experience Zero-Gravity – Luxury Space Tours', and a 'Get Started' call-to-action button."* Once that's in place, use Visual Edit to adjust imagery or text as needed ²⁸⁴ . Next prompt: *"Below that, add a Features section with three columns highlighting: 'Zero-Gravity Flights', 'Orbital Stays', 'Personal Astronaut Training' – with icons."* And so on for pricing, FAQ, contact. This allows more control and ensures each section is as desired, at the cost of a few extra prompts. Many builders prefer this to avoid the AI skipping or mashing sections.
- **Visual & Interactive Polish:** After the basic content is there, refine the design:
 - Use **Visual Editor** to fine-tune spacing, alignment, and images (maybe replace placeholders with actual images from your assets or via Unsplash/Pexels) ²⁸⁵ ²⁸⁶ .
 - Prompt for design tweaks: *"Apply a dark theme with a starfield background in the hero. Make the CTA button glow."* The AI can adjust Tailwind classes for theme consistency ²⁸⁷ .

- Add **animations or effects** for extra “razzle dazzle” ²⁸⁸ ²⁸⁹ . For example: “*Animate the hero text to fade in, and make a small rocket image slowly drift across the screen (using CSS or canvas).*” Lovable can incorporate small animations with Tailwind classes or P5.js for canvas effects ²⁹⁰ . Use these sparingly to avoid distraction, but a bit of motion can make the page feel premium.
- The blueprint example “HorseX Live” (a humorous space-horses theme) incorporated:
 - A dark space theme with stars (perhaps using a canvas or CSS background animation) ²⁸⁹ .
 - Animations (orbiting horses or flickering stars) using P5.js sketches embedded in the page ²⁹⁰ .
 - Logo and favicon generation (AI can help create a simple logo via DALL-E or provide a placeholder) ²⁹¹ ²⁹² .
 - Pricing tables and FAQ toggles (the AI can create an interactive FAQ accordion easily, maybe with a little JS or using HTML `<details>` tags for simplicity).
- For **mobile responsiveness**, toggle the preview to mobile and ensure sections stack nicely. If not, prompt: “*Make the features section responsive: on mobile, stack the three feature cards vertically with some padding.*” The AI will add appropriate Tailwind classes (like `flex-col` on small screens).
- **SEO considerations:** The AI by default may include meta tags if you said “follow SEO best practices” ¹⁷⁸ . Check the page’s `<head>` for a good `<title>` and `<meta description>` . If missing, prompt: “*Add an SEO-optimized title and description: ‘SpaceTours – Luxury Zero-Gravity Flights for Elite Travelers’ and a compelling meta description about space tourism.*” ²⁹³ . Also ensure images have alt text (the AI often puts placeholder alt, you can refine them).
- **Forms (if any):** Many landing pages have a contact or sign-up form. Lovable can wire it to Supabase or email (using Resend) quickly. If you want a simple “contact us” form that emails you, see Resend integration above: one Edge Function call on submit to send an email ²⁹⁴ ²²⁴ . Alternatively, store submissions in a Supabase table for later follow-up.
- **Launch & Iterate:** Once design is good, publish the page (maybe at a custom domain via Lovable or Netlify) ²⁹⁵ . You can share the preview link with colleagues for feedback early too. Landing pages often go through iterations as marketing folks tweak copy or design – with Lovable, you can implement feedback literally in minutes via prompts or visual edits. For instance, if team says “we need a testimonial section with quotes,” just prompt to insert that section with dummy quotes, then fill real text.

In summary, the Landing Page blueprint is about **fast design and polish**. Lovable can produce a near-finished landing page ridiculously fast, leaving you to focus on good copywriting and imagery. Use frameworks like 21st.dev for nice components (the doc suggests copy-pasting or prompting for high-quality Tailwind components for navbars, hero, etc.) ²⁹⁶ . The result: a professional page that would typically take a web designer days, delivered in maybe an hour or two.

7.2 Prototype/MVP Blueprint: When you have an app idea and want to build a working prototype or MVP (Minimum Viable Product) to test it, Lovable excels at generating functionality quickly. The focus here is on **speed and core features**, not perfect polish. Let’s outline building a simple application prototype:

- **Start with a Fresh Project & Supabase Setup:** Let’s say our example is a “Notes app” MVP (users can sign up, create notes).

- Connect Supabase (for auth & DB) early ²⁹⁷ ²⁹⁸ . In Supabase, enable email/password login (disable email confirmations if you want to avoid setting up email) ²⁹⁹ ³⁰⁰ – the prototype can have simpler auth flows.
- Plan the schema: e.g., a `notes` table (id, user_id, content, created_at). You might do this via Supabase dashboard or by prompting Lovable: “Create a Supabase table ‘notes’ with columns: id (UUID), user_id (UUID reference to auth.users), content (text), created_at (timestamp default now).” ³⁰¹ ³⁰² . The AI can execute that via Supabase’s migration (Edge Function or direct API calls).
- **Scaffold the Project Structure:** Lovable Quickstart suggests creating a blank project and then adding features ³⁰³ ³⁰⁴ .
 - Start with a basic navigation (maybe a nav bar with a Login button or a placeholder).
 - Implement auth first (since many features depend on knowing the user). A prompt like “Add a signup page and login page using Supabase Auth. Provide a NavBar with ‘Login’ and ‘Signup’ links when not logged in, and ‘Logout’ when logged in.” The AI will scaffold forms and use Supabase JS library for authentication ³⁰⁵ ³⁰⁶ .
 - Test authentication early (create a test account via the UI, see that it goes into Supabase).
 - Use Visual Edit to ensure the forms are user-friendly (labels, etc.), since the AI might default to minimal forms.
- **Implement Core CRUD Features One by One:** For the notes app:
 - **Create Note:** Prompt: “After login, navigate user to a Dashboard page that lists their notes. On the Dashboard, include a form with a textarea to create a new note and a ‘Save’ button. When saved, store the note in Supabase (notes table) with the current user’s ID, then display it in the list.” The AI will:
 - Generate a `Dashboard` or `NotesPage` component,
 - Use Supabase from the front-end or an Edge Function to insert the note ³⁰⁷ ³⁰⁵ ,
 - Then fetch or update state to show it. Possibly it’ll use a real-time subscription or just re-fetch after insert.
 - **Read Notes (Listing):** Likely included above: the AI will list notes by pulling from Supabase with `supabase.from('notes').select().eq('user_id', currentUser)`.
 - **Update Note:** You might not need in MVP or can include: “Allow clicking a note to edit it (maybe inline or on a separate page). Implement an edit form that updates the note content.” The AI can do a simple modal or navigate to a note detail page with content editable and save on change.
 - **Delete Note:** “Add a delete button next to each note that removes it after confirmation.” The AI will call `supabase.from('notes').delete().match({id: note.id})` and filter the list out.

Use Chat Mode in between to verify logic if something isn't working. For example, after initial creation, ask: “Are we handling the case where supabase returns an error (like no internet)? If not, let’s add an alert on error.” The AI can add basic error handling.

- **Use Real Data Early:** The tip from docs: “Use Real Validation, Fast” ³⁰⁸ means try out actual flows with realistic data to catch issues. Create multiple notes, test edge: empty content should maybe be disallowed (AI might not do validation by default – prompt for it: “Add a check: don’t allow saving an empty note, show an error message.”). If multi-user, create a second account and ensure they only see their notes (Supabase RLS needs to enforce that; the AI might or might not have enabled RLS in the

rush of MVP creation). Before you consider the prototype done, double-check that – perhaps explicitly: *“Ensure each user can only access their own notes in the database. Suggest how to enforce if not already.”* The AI might instruct adding an RLS policy like `auth.uid() = user_id` on notes.

- **Keep UI simple:** MVPs don’t need fancy styling. Use a basic Tailwind theme (Lovable by default uses a neutral style). If you want, you can apply a premade Tailwind theme or component library quickly (maybe integrate DaisyUI by npm if desired – Lovable can install and use it if asked, since it supports using npm packages ³⁰⁹). But often, default styling is fine at prototype stage.
- **Remix for clean-up if needed:** Sometimes while racing to build the prototype, you or the AI may create some messy code (maybe a component does too much). Once it’s functionally working, you can have the AI refactor: *“Refactor the note listing to a separate component <NoteList> for cleanliness.”* Or if it’s really messy, consider Remix to restructure now that you know what you need.
- **Timeboxing & Avoiding Over-optimization:** The point of an MVP is to validate an idea, so don’t over-engineer. Use Lovable’s speed to your advantage by adding features rapidly, but also know when to say “good enough.” For example, don’t spend hours making the perfect pixel alignment – if it’s usable and clear, move on. You can always refine after getting user feedback. The AI might tempt you to keep adding “one more feature” (because it’s so easy), but focus on the core: for notes, maybe you don’t need tagging or sharing in MVP – skip those until core flows are validated.
- **Onboarding & Demo Data:** If you’ll be showing this prototype, perhaps include some sample data or an easy way to add it (so the app isn’t empty). You can prompt: *“If a new user signs up, show a ‘Getting Started’ message or create a sample note for them automatically.”* The AI can seed a welcome note for new accounts (detect first login via some flag in user profile table, etc.). Small touches like that impress stakeholders.

7.3 SaaS Application Blueprint: Building a Software-as-a-Service app is basically taking a prototype and fleshing it out with production-ready considerations: multi-user, roles, billing, custom domain, security hardening, etc. Lovable can accelerate this too. Let’s outline with an example SaaS: say a “Journaling App” that we’ll offer as a service with free and premium tiers (this loosely follows the docs’ SaaS example of journaling + calorie tracker combined) ³¹⁰ ³¹¹.

- **Plan the Architecture:** In SaaS, planning is key because you’ll maintain it long-term. Chapter 4’s guidelines on system design apply. Outline your pages (maybe a Landing page, a Dashboard for logged in users, a Journal entry page, Analytics page), the data models (entries, user profiles, etc.), and integration needs (if premium features, Stripe for payments; for AI analysis, OpenAI integration, etc.) ³¹⁰ ³¹¹. Feed this plan to Knowledge File so AI is aware.
- **Start Frontend-First (Recommended for SaaS):** As docs said, begin with UI scaffolding ³¹² ³¹³:
- **Outline pages and flow via prompt:** e.g., *“Create an outline of the app: A Landing page (marketing), a Sign Up page, a Login page, a Dashboard with sidebar (sections: Journal, Nutrition, Settings), and a Chat page for AI Q&A.”* The AI might not generate them all in one go, but it helps to break it into tasks:
 - Build the main app layout with a sidebar (just placeholder links) ³¹⁰.
 - Add content pages one by one (Journal, Tracker, etc.), possibly with dummy content first.

- **Screenshots or wireframes:** If you have a specific design in mind, you can even draw a quick wireframe and show it to AI with *“Match this layout.”* Or find an inspiration site and describe it (or use a “Figma to Lovable” if you have a Figma design – Lovable can import from Figma plugin, according to docs).
- Aim to get a prototype UI where navigation works (even if data is dummy) before adding real logic. This ensures the UX is right.
- **Connect Backend & Auth:** Once UI is set:
 - Hook up Supabase for database and auth ³¹⁴ ³¹⁵ . Or if you prefer an external auth like Clerk, integrate that (Clerk might be good if you plan to allow SSO, etc., in a SaaS context).
 - Define data tables: e.g., for journaling, `entries` table with mood, energy, text; for calorie tracker, maybe `meals` table with items and calories. The docs example had “journaling and calorie tracker” with some overlapping concepts (like analyzing mood and meals) ³¹⁶ ³¹⁷ .
 - Ensure multi-tenancy if relevant: If each user’s data is separate, RLS in Supabase covers it. If you have “organizations” (B2B SaaS), you might integrate Clerk Orgs or have a `team_id` on data with RLS by team. Plan that early so the AI incorporates `team_id` in data model.
- **Implement Core Features with an Eye on Quality:**
 - Use the incremental approach from prototype but now consider edge cases and robustness. For example, journaling app:
 - Add ability to rate each day’s mood (1-5) and write a note ³¹⁸ .
 - Add ability to upload a photo (maybe to Supabase Storage) for each entry.
 - Ensure the UI for past entries (list by date) is paginated or at least not going to break with lots of data.
 - If including AI analysis (like “AI, how was my week?”), integrate OpenAI via an Edge Function now, with error handling for API failures ³¹⁹ ³²⁰ .
 - Premium features: e.g., premium users get a “personal AI coach” or the calorie tracker tab. Use feature flags (like `user.role` or a `is_premium` boolean in user profile) in the code to show/hide features ³²¹ ³²² .
- **Integrate Payments (Stripe) Early:** For SaaS, you’ll likely have subscription plans. Use Lovable’s Stripe integration as described in Chapter 5:
 - Decide on plans (Free, Pro, etc.), set them up in Stripe, get price IDs.
 - Prompt Lovable to integrate Stripe checkout and portal ³²¹ ³²² . E.g., *“Integrate Stripe: free plan gets 5 entries/day, premium (\$10/mo) gets unlimited + AI analysis. Implement upgrade flow: if user not premium, show ‘Upgrade’ button linking to Stripe checkout for Premium. After payment, mark user as premium and unlock features.”* The AI will handle much of this logic, including edge cases like if payment fails (maybe it will use Stripe’s built-in UI messages).
 - Test in Stripe test mode vigorously (subscribe, cancel, etc.) because that’s core to SaaS business logic.

- Make sure to secure any content that should be premium-only both in UI (don't show button) and backend (if a free user tries an API call to premium feature, have the Edge Function check their role and refuse if not allowed).
- **Polish and Harden:** Now treat it like a real app:
 - Add form validations (e.g., journal entry can't be empty, meal input must be positive number for calories).
 - Consider performance: if each user potentially logs hundreds of entries, implement lazy loading or limit initial query (maybe show last 7 days by default, with "load more").
 - **Security audit:** Re-run the security checklist: keys secure, RLS in place (especially if multi-tenant), no sensitive info in front-end, etc. Possibly have AI do a security scan as mentioned before launching.
 - **SEO & PWA:** If the SaaS has a public front (like a marketing site), optimize SEO. If it's mostly an app behind login, SEO not critical except maybe your landing page. Consider making it a PWA if it's the kind of app users might use daily on mobile (e.g., journaling each morning – a PWA would let them open it like a native app).
 - **Beta Testing:** Before fully launching, perhaps invite some testers. Use Lovable's *Launched* listing or just share the app link. Gather feedback and use the agility of Lovable to iterate. For instance, if testers say "the UI is confusing on the analytics page," you can fix it literally in minutes with AI. This is a huge advantage – incorporate feedback extremely fast, which is golden in early SaaS development.
 - **Scaling Considerations:** If your SaaS grows, code might need optimization (the AI's code is not always the most efficient). Keep an eye on any slow queries or heavy functions. You can always come back and prompt the AI to optimize specific parts (as done in Chapter 4 with performance audit). Also, if needed, you can eject from Lovable partially by editing code in Dev Mode manually or in GitHub for fine control.

The SaaS blueprint is essentially an amalgamation of everything: design, integrations, best practices, deployment. Lovable shortens the time to go from idea to a revenue-capable product. By following the structured approach (UI -> backend -> features -> payments -> polish), you ensure no step is overlooked.

7.4 Other Use Cases and Resources: The above cover common ones, but Lovable can handle more: - **Mobile Apps:** While Lovable output is web, you can frame mobile experiences. E.g., build a mobile-optimized web app, then wrap with Capacitor. The blueprint would be similar to SaaS but focusing on responsive/mobile design from the start (maybe using a mobile UI kit). - **Internal Tools/CRUD apps:** If you need an admin panel or internal tool, Lovable is superb. Treat it like SaaS minus the marketing site and payments: focus on CRUD features, maybe integrate with things like CSV export or external APIs for data. The blueprint would emphasize using available npm packages (Lovable supports adding packages ³²³, so for admin maybe use something like Chart.js for charts or Tabulator for tables – just prompt to install and use). - **Browser Extensions & Niche:** While Lovable builds web apps, you could create a tool then embed it in a Chrome extension (maybe using the extension to open a panel with your Lovable app UI). The "Prompt2MVP" extension concept in docs suggests capturing screenshot -> prompt, which is a different angle (for idea inspiration). - **Game prototypes:** Yes, you can even do basic HTML5 games. The AI can use p5.js or Three.js as per integration list ³²⁴ ³²⁵. It might not replace a game engine for complex games, but for simple interactive visuals or edu-games, give it a shot as a fun exercise.

To wrap up: **use case blueprints** are not strict rules but proven patterns. The key is to leverage Lovable's strengths (fast UI, quick integration, iterative improvement) and mitigate its weaknesses (be explicit to avoid wrong assumptions, check its work for quality, add human judgment where needed). By following these blueprints and cross-linking relevant Lovable features (like we did, e.g., adding Clerk to a SaaS for robust auth, using Supabase for backend, etc.), you can build just about anything.

Each blueprint also cross-pollinates: e.g., a SaaS will need a landing page too (so Chapter 7.1 applies inside 7.3). A prototype might become a SaaS. So mix and match as needed.

At this point, you should feel equipped to plan and create projects of various scales with Lovable. In the final chapter (Appendix), we'll provide some prompt templates, checklists, and a glossary of terms as a handy reference as you embark on your AI-assisted development journey.

8. Appendix: Prompt Templates, Checklists, and Glossary

This appendix provides quick-reference materials: example prompt templates for common scenarios, a consolidated checklist to review before launching your project, and a glossary of Lovable-specific terms and development concepts for clarity.

8.1 Prompt Templates Library: Use these patterns to speed up writing prompts (feel free to modify details to fit your project):

- **Feature Kickoff (Structured):** When starting a new feature or page, use a structured prompt for clarity:

```
**Context:** You are a Lovable AI building a <FeatureName> in a <ProjectType> app.
**Task:** Implement <feature description>.
**Guidelines:**
  - Use <specific libraries or tech> (e.g., Tailwind for styling, Supabase for data).
  - Maintain existing design style (e.g., dark theme, mobile-responsive).
**Constraints:**
  - Do not modify <critical file/component> other than adding imports.
  - Ensure no breaking changes to current functionality.
```

Example usage: "Implement **Profile Picture Upload**. Context: logged-in user on settings page. Task: allow user to upload a profile image and save it. Guidelines: use Supabase Storage for image, update user's `avatar_url` in DB, show preview. Constraints: don't affect other settings fields, image max 5MB."

This yields a focused result with an upload button, using Supabase securely [326](#) [327](#) .

- **UI/UX Enhancement:** When refining design without altering logic:

The app UI should be improved, ****without changing any functionality****:

- Keep all existing logic and IDs.
- ****Visual Enhancements:**** <list changes> (e.g., "Restyle dashboard cards with a modern card design, add subtle shadow, improve contrast on text").
- Ensure the app behaves exactly as before (no JS changes).

Example usage: "Polish the landing page UI: use a gradient background for hero, animate the CTA button on hover, and increase font sizes for headings for better readability 328 329 ." This instructs the AI to only tweak styles, not logic.

• **Locking Scope (Do/Don't):** To focus the AI on a specific area or prevent collateral edits:

Modify the ``<ComponentName>`` component to achieve <goal>, and ****do not change any other files or components****.

- Do not alter the behavior of <X> (e.g., "Login flow").
- Focus only on <Y> (e.g., "the styling of this component").

Example: "In `Header.jsx`, add a navigation menu icon for mobile. **Do not edit** any routing or other components 35 ." This ensures the AI confines changes to Header.jsx, reducing unintended side effects.

• **Bug Investigation (Chat mode style):**

"We're encountering [Error/Issue] when [action]. Let's diagnose this:

1. What is likely causing the error?
2. What parts of the code are involved?
3. Suggest a plan to fix without directly coding yet."

Example: "After adding the note feature, saving notes sometimes fails with `403 Forbidden`. Let's diagnose: 1. Why might a Supabase insert return 403? (Perhaps RLS denying?). 2. The code posting the note and DB policy. 3. Plan to fix (maybe adjust RLS or use the correct user ID) 95 100 ."

This yields an analytical answer pinpointing the root cause (like missing auth) and a fix strategy, which you can then implement.

• **Code Explanation (Reverse Meta):**

"Explain the purpose of the following code and how it works:

```
```js
<paste code or function>
```

- Describe inputs, outputs, side effects. - Mention any edge cases handled or not handled.

This prompt uses backticks to include code (which Lovable's AI will read) and asks for a summary <sup>51</sup>. Good for understanding AI-generated code or for adding documentation.

*Example:* Ask it to explain an Edge Function for Stripe payments it created <sup>208</sup>. The AI will produce: "This function `createCheckoutSession` takes a price ID and user ID, creates a Stripe Checkout Session for that price and returns the URL... It assumes the user is logged in (uses their ID), but it doesn't handle errors if price is invalid..."

These templates can be combined. For instance, you might lock scope and request UI enhancements in one prompt for a specific component. Always adapt wording to your app's context.

**8.2 Pre-Launch Checklist:** Before considering your project "done" or ready for users, run through this list:

- **Functionality:**

- Have all core user stories been implemented and tested (including edge cases)? <sup>80</sup> <sup>330</sup>
- Does every button, link, and form input do what it's supposed to (on all devices)? Test thoroughly.
- Are error states handled gracefully (e.g., network errors show a message, invalid inputs show validation feedback)? <sup>35</sup> <sup>73</sup>

- **Design & UX:**

- Is the app **responsive** on different screen sizes (mobile, tablet, desktop)? Use Lovable's preview toggle and adjust layout if needed <sup>285</sup> <sup>331</sup>.
- Are interactive elements clearly visible and accessible (buttons large enough, text readable with sufficient contrast)? <sup>280</sup> <sup>281</sup>.
- **Accessibility:** Is semantic HTML used (e.g., forms have labels)? Are images with meaningful content given `alt` text? Test navigating via keyboard only or using a screen reader on key flows <sup>280</sup> <sup>281</sup>.
- **Consistency:** Do pages share a common style guide (colors, typography)? If not, prompt the AI to standardize (e.g., "Use consistent heading styles across pages").

- **Performance:**

- Do pages load quickly? Check if any images are too large (compress them) <sup>137</sup> <sup>138</sup>.
- If using third-party scripts or heavy libraries, ensure they're only loaded where needed (the AI might include all in bundle; consider code-splitting if app is large).
- Run a Lighthouse or similar audit for performance and fix obvious slow points (the AI can assist optimizing code or queries) <sup>135</sup> <sup>332</sup>.

- **Security:**

- All API keys and secrets are stored securely (Supabase secrets, Lovable Add API Key, etc.), not exposed in client code [140](#) [13](#) .
- Supabase RLS policies are in place for any sensitive data tables [333](#) [147](#) . Test by trying to access data as a different user – ensure it fails.
- Validate inputs on server-side too if applicable (Edge Functions double-check data from client). The AI often does basic validation client-side, but adding server checks is wise for production.
- The Lovable Security Scan shows **no warnings** on publish [147](#) [153](#) . If any warnings remain, address them (even if it's “just a warning”).
- If file uploads are allowed, ensure file type/size restrictions to prevent abuse (AI can add these checks if not present) [334](#) [335](#) .

#### • Integrations:

- Stripe: run test purchases, upgrades, downgrades. Ensure webhooks (or polling) update user status properly [336](#) [322](#) . Check that non-paying users cannot access paywalled features (try using a test account).
- Email (Resend): send test emails and verify deliverability (check spam folder, etc.) [337](#) [224](#) .
- External APIs (OpenAI, etc.): handle rate limits or errors gracefully (e.g., show “AI is busy, try again” if API returns 429 or fails) [234](#) [235](#) . Also monitor usage to avoid unexpected costs.

#### • Content & SEO:

- Update any placeholder text (the AI may have left lorem ipsum or generic content) – replace with real copy that you want.
- Set appropriate **Page Titles & Meta Descriptions** for all major pages (particularly landing page) [278](#) [338](#) .
- Generate an XML Sitemap if SEO is relevant (the AI can do this on request) [179](#) [339](#) .
- Add a `robots.txt` if necessary (AI can generate one to allow or disallow certain paths) [340](#) [341](#) .
- If it's a web app behind login, consider adding basic SEO content to your landing page only (maybe a blog or docs for marketing, which Lovable can also generate scaffolding for if needed).

#### • Deployment:

- If using a custom domain, verify SSL and domain setup (Lovable/Netlify/Vercel will usually handle SSL via Let's Encrypt – ensure it's active by visiting `https://yourdomain`).
- Test the app in production environment (some bugs only appear when deployed, like environment variable issues or case-sensitive path issues).
- Enable analytics (if wanted) by adding e.g., a Google Analytics script or similar (the AI can embed an analytics snippet if you prompt it).
- Prepare for scaling: e.g., increase Supabase table Row Limits if needed, or move to a higher tier if expecting many users (not a development step per se, but good to note).

#### • Backup & Handoff:

- Ensure your GitHub repo is up to date with latest code (Lovable does this if integrated, but double-check).
- Write minimal documentation for future devs (and yourself): The Knowledge file doubles as a good project README at this point <sup>117</sup> <sup>118</sup> . Update it to reflect final state (features list, how to run/deploy, etc.). You can even ask AI to create a README.md summarizing the project – it will use the knowledge file and code to produce one.
- Pin the final version in Lovable (so you have a restore point if needed) <sup>115</sup> .

#### • Community & Support:

- If launching publicly, have a plan for support/feedback (maybe a Discord or simply an email contact). Perhaps integrate a simple feedback form or chat widget if appropriate.
- Consider posting on Lovable’s “Launched” page or community showcase – it’s a good way to get initial users and feedback <sup>163</sup> <sup>165</sup> .

This checklist may seem long, but thanks to Lovable, many items are either auto-handled or quick to fix. Run through it systematically – it’s normal in development cycles to spend time on this polish phase.

### 8.3 Glossary of Terms:

Below are definitions of terms you’ll encounter in Lovable development (many are clickable in Lovable’s own docs):

- **Prompt:** A textual instruction you give to the AI to perform a task <sup>342</sup> . In Lovable, prompts drive code generation or modifications – essentially your way of “coding” in natural language. Writing effective prompts (clear, detailed) is crucial for good results.
- **Chat Mode:** An interactive mode in Lovable where the AI acts as a conversational assistant rather than immediately editing code <sup>343</sup> . Use Chat Mode to debug, brainstorm, or get explanations – it’s like talking with a pair programmer. Changes are not applied to your project unless you explicitly ask within Chat Mode to do so (or switch back to edit mode).
- **Edit Mode (Default Mode):** The normal mode where your prompt leads directly to code changes in the project. When you type a prompt with the project editor in focus, that’s Edit Mode (the AI will usually reply with code modifications). Edit Mode is where your project’s code is actually altered, and it consumes your “edit” credits.
- **Visual Edit (Visual Editor):** A mode where you can make changes by clicking and dragging elements on a live preview of your app <sup>62</sup> . It generates AI prompts under the hood for styling/HTML changes, but you don’t type them – you use GUI controls. It’s great for quick UI tweaks without spending prompt credits.
- **Dev Mode:** A feature that lets you directly view and edit the project’s code (like in a VS Code editor) within Lovable <sup>8</sup> . Only available to paid plan users, it’s useful for manual adjustments, reading code, or making bulk changes that might be tricky via prompt. Dev Mode syncs with AI mode – you can still use prompts, but you have the freedom to hand-edit too.



- **Knowledge File:** A markdown file (`knowledge.md`) in your project that stores project-specific context <sup>344</sup> <sup>345</sup>. Lovable automatically includes it with every prompt to give the AI more knowledge about your app (essentially your app's wiki/brain). Populate it with requirements, data schemas, and design decisions. It helps maintain consistency and reduces repeating yourself in prompts.
- **Supabase:** An open-source backend as a service (BaaS) providing Postgres database, authentication, storage, and serverless functions <sup>86</sup>. Lovable integrates with Supabase to give your app a persistence layer quickly. Think of Supabase as the “database and auth server” for your Lovable app. Key concepts: tables (data storage), RLS (security rules), Edge Functions (serverless code).
- **Edge Function (Supabase Edge Function):** A serverless function (written in TypeScript/JavaScript) that runs on Supabase's servers, typically used to handle secure or backend logic (like calling third-party APIs with a secret) <sup>151</sup> <sup>234</sup>. Lovable can create and deploy these as part of integration (for example, a Stripe webhook handler or an OpenAI call). They allow you to keep secrets safe and execute code outside the user's device.
- **Row-Level Security (RLS):** Supabase feature to restrict which rows of a table a certain user can access <sup>333</sup>. For multi-user apps, you enable RLS and write policies (SQL rules) that e.g., allow “users can select their own data where `user_id = auth.uid()`”. Lovable flags if RLS is off on a table and likely should be on (it warns on publish) <sup>146</sup>. Always review RLS for any sensitive data tables before launch.
- **Clerk:** A third-party service for user management (authentication, user profiles, teams) with pre-built UI <sup>209</sup>. Lovable can integrate with Clerk as an alternative to Supabase Auth, giving a richer auth experience (social logins, etc.). If you see references to Clerk, it usually means a more advanced auth setup is in use.
- **Resend:** A service for sending emails via API <sup>346</sup>. In Lovable context, Resend integration means your app can send emails (welcome emails, notifications) using Resend's API. Often done via an Edge Function to keep API keys secret.
- **Make (formerly Integromat):** A no-code automation platform. Lovable's “Integration with Make” refers to connecting your app to Make to orchestrate complex workflows across apps (like sending Slack messages when something happens, etc.) <sup>228</sup>. Essentially webhooks and API calls bridging Lovable and other services through Make's visual flows.
- **GitHub Integration:** The linking of your Lovable project to a GitHub repo for version control <sup>9</sup> <sup>171</sup>. Once integrated, every AI change is a commit, and you can collaborate via Git as normal. It brings transparency and backup to your AI coding. If you see “real-time sync with GitHub” mentioned, that's this integration.
- **Messaging Limits (Credits):** Lovable's plans limit how many messages/edits you can do (because each prompt uses OpenAI/AI calls) <sup>347</sup> <sup>348</sup>. Free tier gives some daily credits, Pro/Team have monthly quotas. “Messages” usually count both chat and edit prompts (with possibly different weights). Keep an eye on usage; use chat for thinking to save edit credits, as discussed.

- **Remix:** A feature to duplicate a project (either your own or someone else's if public) and start a new independent version <sup>90</sup> . Useful for forking an example, or resetting your project with current state as baseline to try different approaches without risking the original. A remix does not carry over environment secrets and might disconnect Supabase for safety, so you re-connect if needed <sup>102</sup> .
- **Pin/Version History:** Lovable automatically keeps versions (commits) of your project as you go. You can "Pin" a version to label it (like v1.0 launch) <sup>115</sup> . Version history lets you roll back if needed. It's essentially a simplified view of your Git history (or internal history if no GitHub).
- **Mintlify (Powered by Mintlify):** Not a Lovable feature, but you might see it in Lovable docs (like "Powered by Mintlify" at bottom) <sup>349</sup> <sup>350</sup> . Mintlify is just the documentation framework Lovable uses. Not directly relevant to building your app, except that it's the reason Lovable's docs have that layout. You don't need to interact with it.
- **MVP (Minimum Viable Product):** A product with just enough features to be usable by early customers, used for validating the idea. In Lovable terms, it's what you build initially to test the waters (the "Prototype" we discussed) <sup>297</sup> .
- **CRUD:** Create, Read, Update, Delete – the basic operations on data. Many Lovable-generated apps revolve around CRUD functionality (like our notes or journal entries). The AI is quite good at scaffolding CRUD forms and lists quickly.
- **Inspiration (Templates/Examples):** Lovable has example use-case apps and templates (like "Simple CRM", "Landing Page", etc.) mentioned in docs <sup>351</sup> <sup>352</sup> . You can remix these or draw inspiration. For instance, if stuck on how to implement something, recalling an "Inspiration time" example from docs might help the AI (or you) find a solution pattern.

That covers the key terms. Knowing these will help you navigate Lovable's interface and understand documentation or AI's responses better.

With this guide, you have a comprehensive reference to build and refine applications using Lovable. The combination of structured best practices and AI's power can make you a "lovable" developer – one who delivers solid results quickly and enjoys the process. Happy building!

**[Guide compiled from Lovable documentation and examples – for more, see [Lovable.dev/docs](https://docs.lovable.dev/docs)]**

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<sup>1</sup> <sup>2</sup> <sup>3</sup> <sup>4</sup> <sup>5</sup> <sup>163</sup> <sup>252</sup> <sup>274</sup> <sup>275</sup> Welcome - Lovable Documentation

<https://docs.lovable.dev/introduction>

<sup>6</sup> <sup>7</sup> <sup>27</sup> <sup>28</sup> <sup>29</sup> <sup>30</sup> <sup>31</sup> <sup>32</sup> <sup>33</sup> <sup>34</sup> <sup>36</sup> <sup>37</sup> <sup>38</sup> <sup>39</sup> <sup>40</sup> <sup>41</sup> <sup>42</sup> <sup>43</sup> <sup>44</sup> <sup>45</sup> <sup>46</sup> <sup>47</sup> <sup>48</sup> <sup>49</sup> <sup>50</sup> <sup>52</sup> <sup>56</sup> <sup>57</sup> <sup>58</sup> <sup>59</sup>  
<sup>60</sup> <sup>61</sup> <sup>63</sup> <sup>64</sup> <sup>65</sup> <sup>66</sup> <sup>67</sup> <sup>70</sup> <sup>71</sup> <sup>72</sup> <sup>74</sup> <sup>75</sup> <sup>107</sup> <sup>280</sup> <sup>281</sup> Prompting 1.1 - Lovable Documentation

<https://docs.lovable.dev/tips-tricks/prompting-one>

<sup>8</sup> Dev Mode - Lovable Documentation

<https://docs.lovable.dev/features/dev-mode>

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